

Safe designs, unsafe ecosystems: deployment-layer selection in an evolutionary AI race

Author One¹ and Author Two^{1,2}

¹Affiliation One

²Affiliation Two

Abstract

We study evolutionary dynamics in which the payoff earned in the game and the payoff that governs replication are different functions of the same interaction. The separation is the defining feature of an ecosystem of deployed AI agents: an agent acts in a competitive race, but whether its *design* is replicated is decided by a principal who does not play that race and who bears only a fraction λ of the external harm the design causes. We call the difference between the two functionals the selection wedge, embed it in the reduced four-strategy AI race of Fernández Domingos and Han, and analyse the resulting replicator and finite-population dynamics. The wedge collapses to a single control parameter, the effective liability $L = \lambda h$, and every invasion condition is affine in L , which yields closed-form thresholds. Four consequences follow. First, a conditionally unsafe design weakly dominates the unconditionally unsafe one at every L , so a pre-deployment filter that scores designs against a safe environment removes only a weakly dominated design and, under a common start measure, leaves the long-run unsafe frequency unchanged. Second, unconditional and conditional safety are exact payoff twins, so the invasion barrier of the safe face is set by a neutrally drifting composition variable, and it falls by a factor of 77.7 across that face. Third, long-run harm is *not monotone* in liability: on $L \in (4.27, 42.77)$ liability taxes the retaliation that conditional safety relies on, the unsafe coexistence becomes uninvadable, and crossing the lower edge from below lifts the basin-averaged unsafe frequency from 0 to 0.32. A window of this kind exists at every prize and private-risk level we examine, with multiplicative width between 4.2 and 12.8, and in a finite population the unsafe attractor survives between 65 and 3.9×10^{13} generations. Fourth, coupling the dynamics to a non-decreasing stock of diffused capability that erodes enforcement produces hysteresis whose width is exactly computable and depends on the erosion channel only through the enforcement that survives full diffusion, not on the speed of diffusion. We then test how far the mechanisms travel: they are inherited by any selection dynamics driven by the sign of a payoff difference, they survive non-linear liability schedules and richer conditional designs, and the valley closes only when execution error exceeds about 7.5% – at which point the protected state has stopped being safe. Two features of the environment then move the valley in opposite directions by computable amounts. Assortative matching shrinks it entirely from above, because the lower edge is exactly independent of assortment, and closes it at an assortment of 0.354. Booking a fraction q of the blame for an interaction against the wrong party widens it as $1/(1 - q)$, because the design that a first strike exploits is by construction the one with a spotless record; every other measurement error we consider, including zero-mean noise, flat charges and systematic under-reporting, is absorbed exactly. The results locate the governance problem at the selection layer rather than at the design layer, and identify liability regimes that are worse than either extreme.

Keywords: evolutionary game theory; replicator dynamics; technological race; bistability; hysteresis; liability; AI safety

1 Introduction

Evolutionary game theory usually identifies two things that need not be the same: the payoff an individual earns in the game, and the quantity that governs how often that individual is copied [1–3]. Under that identification every standard result – the correspondence between rest points and Nash equilibria, the stability of evolutionarily stable strategies, the invasion analysis of finite populations – is a statement about one payoff matrix used twice.

Ecosystems of deployed AI agents break the identification in a specific way. An agent negotiates, bids, schedules or races on behalf of a principal, and earns a payoff by doing so. Whether the agent’s *design* – its model, its system prompt, its scaffold – is replicated across further deployments is not decided by the agent, and is not decided by the payoff the agent earned in the abstract. It is decided by the principal who deployed it, on the basis of what the principal received. The two quantities differ whenever the design imposes costs on third parties that the principal does not bear. The replicating unit is therefore selected by an agent that never plays the game, according to a functional that is not the social one. Selection here is artificial rather than natural, in the breeder’s sense: the copied variant is the one that served the breeder.

The consequences of that decoupling are the subject of this paper. They are not the familiar observation that individual incentives can diverge from collective welfare, which has been understood since Hardin’s account of the commons [4] and is the content of every social dilemma. The point is structural and sharper: the dynamical system that describes the deployed population is the replicator equation of a game that nobody plays. Its rest points are the equilibria of the principals’ selection game, and the social game enters only as a passive observable. Interventions therefore have very different effects depending on which of the two objects they touch. An intervention on the design pool – safety training, capability evaluations, release gating – changes the set of available strategies. An intervention on the selection functional – liability, disclosure, procurement rules – changes the vector field itself. We show that the first class can be exactly neutral while the second class is not, and that the second class can be non-monotone in a way that makes intermediate regimes worse than none.

The separation itself is not new. The indirect evolutionary approach [5–8] studies populations in which behaviour follows subjective preferences while selection operates on material payoffs, and Heifetz et al. [9] show that in almost every game some distortion of a player’s true payoff is beneficial and survives payoff-monotone selection. Our construction is the mirror image. There the distortion sits in the behaviour and is selected on the socially correct functional, and it is endogenous; here behaviour follows the material race payoff, the distortion sits in the *selection* functional, and it is an exogenous policy parameter. What is new is not the existence of two functionals but the direction of the wedge and its consequences for governance instruments, which is why our results are statements about interventions rather than about which preferences survive.

Concern about selection pressure over deployed AI systems has been raised informally [10] and appears as one of seven risk factors in a recent multi-agent risk taxonomy [11], but we are not aware of a formal treatment. Formal analyses of AI races have instead studied the race itself: whether speed dominates safety [12], how regulation and heterogeneity change the prevalence

of unsafe development [13, 14], and how commitments and delegation alter outcomes [15–17]. Those models keep the classical identification: the racing actors are the reproducing actors. We keep their interaction layer and change the selection layer.

Contribution. We formalise deployment-layer selection, embed it in the reduced AI-race game of Fernández Domingos and Han [18], and derive its consequences analytically and numerically. Section 2 specifies the two layers and the numerical protocol, Section 3 contains the analysis, Section 4 draws out what the results imply for governance, and Section 5 concludes. The four results are:

1. **Solo evaluation cannot move the equilibrium** (Section 3.2). The conditionally antisocial design CAS weakly dominates the unconditionally unsafe design AU at every $L \geq 0$. A filter that rejects designs whose unsafe frequency against a safe environment exceeds ε rejects AU and admits CAS, whose self-play unsafe frequency is 1. The only attractor the filter deletes is one of maximal harm that the surviving pool reproduces.
2. **The safe face is neutral and its barrier is not** (Section 3.3). Always-safe and conditionally-safe designs are exact payoff twins, so the safe face carries no selection gradient. Its invasion barrier nonetheless falls from $L^* = 42.77$ at the unconditional vertex to $L^* = 0.55$ at the conditional vertex, a factor of 77.7; protection is governed by a variable that drifts.
3. **Long-run harm is non-monotone in liability** (Section 3.4). On $L \in (4.27, 42.77)$ the interior rest point of the unsafe face becomes uninvadable by conditional safety, because liability also taxes retaliation. Crossing the lower edge from below raises the basin-averaged unsafe frequency discontinuously from 0 to 0.32. The mixture-independence of the invasion criterion behind this result (Lemma 16) holds for any symmetric matrix game.
4. **Irreversible diffusion makes the transition hysteretic** (Section 3.10). With a monotone environmental state that erodes enforcement, the liability at which a safe regime is recovered exceeds the liability at which it was lost by a factor that is exactly computable and independent of the diffusion rate.

Four limitations should be stated before the results rather than after them. The interaction layer is the deliberately stylised two-player race of Fernández Domingos and Han [18] with four reduced strategies; it abstracts away many actors, gradual capability, and learning. The external harm h is a free parameter with no calibration, so we report the weight-free unsafe frequency as the primary observable, characterise the entire L -axis rather than committing to a point estimate, and normalise thresholds by the race prize. The population is well mixed, so network reciprocity plays no role in the baseline. And the principal is assumed to observe the externality it charges for, which no incident-reporting regime does.

Because those four choices carry the results, three sections relax them one at a time and report which conclusions move and by how much. Section 3.7 adds execution noise and four further conditional designs, and gives the rate at which exact twinning breaks. Section 3.8 replaces the linear liability charge by four other shapes, the replicator equation by three other dynamics, and random matching by assortative matching, which settles what population structure does to the non-monotonicity: it shrinks it, entirely from above, and closes it at an assortment of 0.354.

Section 3.9 replaces the true externality by a measured one and separates the measurement errors that a selection functional absorbs exactly – zero-mean noise, flat charges, and any error that depends only on the counterparty – from the one that it does not, which is booking an incident against the wrong party. That single error is the one that widens the dangerous range without bound. Readers who want to know in advance which claims are general and which belong to one parameterisation will find them sorted in Table 3.

1.1 Related work

Evolutionary models of AI races. Armstrong et al. [12] introduced a formal race model in which information about rivals changes the willingness to skip safety precautions. Han et al. [13] embedded a race in an evolutionary dynamic and showed that whether regulation is beneficial depends on the speed advantage of unsafe development and on the time horizon; Cimpéanu et al. [14] extended the analysis to heterogeneous populations. These models establish that competitive pressure alone can sustain unsafe development. They also share the modelling choice we depart from: the racing actor and the reproducing actor coincide, so the population dynamic is driven by the racing payoff itself. Our contribution is orthogonal rather than corrective – we retain their race and ask what changes when the replication decision is made by a third party.

Two-level selection. A second literature separates the level at which behaviour is expressed from the level at which it is selected. Multilevel selection [19–21] lets a trait that is disfavoured within a group be favoured between groups, so the population dynamics is governed by a composite of two selection pressures. Our construction is a degenerate but sharper case of the same idea: there is exactly one selecting level, it is not the level that plays, and the functional it selects on is a policy variable rather than an aggregate of the lower level. Where multilevel selection asks which of two competing gradients wins, we ask what happens when the lower-level gradient is absent entirely, which is why our comparative statics are in the instrument L rather than in a group-size or migration parameter.

Incentive design in evolutionary games. A related strand designs institutional rewards and punishments to steer an evolving population, and asks what they cost [22, 23]. Those instruments are added to the payoffs of the players themselves and are typically monotone in their budget. The instrument here acts one level up, on a principal who is not a player, and Section 3.4 shows that it is not monotone. The inertness criterion of Proposition 6 is the design-relevant statement in that literature’s language: it says which perturbations of the payoff matrix a designer can make for free, and by contraposition which ones necessarily bite.

Selection on an imperfect measure. The wedge has an older counterpart in contract theory. When a principal cannot observe the action it wants to reward, it must contract on a signal, and the value of a signal is its informativeness about the action rather than its correlation with output [24]. When the signal captures some dimensions of effort better than others, the agent reallocates effort towards what is measured, and the resulting distortion can make a high-powered incentive worse than none [25, 26]. The selection functional here is exactly such a distorted measure, and the mechanism is the multitask one at a different level: liability measures unsafe actions without distinguishing initiation from retaliation, so raising its power reallocates

the population towards designs that do not retaliate. Two things are different. The reallocation is evolutionary rather than deliberate – no agent responds to the incentive, the population composition does – so the distortion appears without any optimisation to blame it on. And the second-best remedies of contract theory, which trade off the power of the incentive against the distortion, have a counterpart here that the static theory does not predict: because the distortion is bistable, the loss is discontinuous in the power of the instrument rather than smooth, and there is a range of intermediate powers that is worse than both endpoints.

Delegation and artificial agents. Fernández Domingos et al. [16] showed experimentally that delegating a collective-risk decision to an artificial agent increases prosocial behaviour, and Terrucha et al. [17] showed that committing to an imperfect delegate can beat acting directly. That literature treats delegation as a commitment device chosen once by a player inside the game. Here the principal is outside the game and acts repeatedly on a population of designs; the object of interest is not the delegate’s behaviour but the selection pressure the principal exerts on the design distribution.

Behavioural evidence on the race. The interaction layer we adopt is the framed experiment of Fernández Domingos and Han [18], in which paired participants repeatedly chose between safe and unsafe development under an uncertain horizon. Its central finding is that unsafe choices track the evolving strategic state of the race rather than elicited risk preferences. That finding motivates the conditional strategies CS and CAS, which carry all of our results. Conditional reciprocation of this kind is the dominant behavioural regularity in repeated social dilemmas more broadly [27–29].

Environmental feedback in evolutionary games. Coupling replicator dynamics to a slow environmental state produces oscillations and bistability [30, 31], and abrupt regime shifts with hysteresis are a well-studied feature of ecosystems with such feedbacks [32]. Existing game–environment models take the environment to be renewable: it recovers when exploitation stops. Deployed capability does not. Our ratchet is the degenerate but practically important case of a state that is monotone by construction, which converts a reversible parameter change into an irreversible regime change.

Liability. The economic analysis of accident law asks when strict liability outperforms negligence and when liability should be combined with regulation [33, 34]. Both analyses are equilibrium statements about a fixed population of injurers. We ask instead what a liability level does to an evolving distribution of designs. The answer is a qualitatively different phenomenon: a window of intermediate liability that is worse than either bound, because the instrument taxes the retaliation on which conditional safety depends. That taxation of retaliation is a second-order free-rider problem [22, 35] in a liability setting.

2 Model

We now specify the two layers. The interaction layer is taken unchanged from the source study [18] so that our results are attributable to the selection layer alone; the selection layer is new.

2.1 Interaction layer: the reduced AI race

Two agents repeatedly choose $a \in \{S, U\}$, safe or unsafe development. Safe development advances the race by $\sigma(S) = 1$ step and unsafe development by $\sigma(U) = 1.5$ steps. The round payoff matrix, rows indexed by the focal action, is

$$\pi = \begin{pmatrix} 1 & 0.6 \\ 2.4 & 2 \end{pmatrix}. \quad (1)$$

The race lasts at least five rounds and each further round ends it with probability 0.2, so the horizon is $T = 5 + G$ with $G \sim \text{Geom}(0.2) - 1$ and $\mathbb{E}[T] = 9$. The agent with greater cumulative progress receives a prize B ; equal progress splits it. A winning or tying agent suffers a setback with probability

$$q_i(T) = p_r^{\max} \frac{n_i^U(T)}{T}, \quad (2)$$

where $n_i^U(T)$ counts its unsafe choices, and a setback removes its task payoff. The baseline uses the experimental values $B = 100$ and $p_r^{\max} = 0.6$; Section 3.6 varies both, and Appendix B records the other two experimental treatments.

The reduced strategy set is the one inferred from the experiment: **AS** always safe; **AU** always unsafe; **CS** safe in round one, then copying the opponent’s previous action; **CAS** unsafe in round one, then copying.

Remark 1 (Exact evaluation). All four designs are deterministic, so the action path of an ordered pair is fixed once the pair is fixed and the only stochastic primitive is T . We therefore obtain the matchup quantities by exact expectation over the horizon law rather than by the Monte Carlo procedure used in the source study [18]. Exactness matters here because several of our thresholds are ratios of small payoff differences, so sampling noise of order 10^{-2} in the payoff matrix would propagate to the third significant digit of those ratios.

With the evaluation of Remark 1 in hand, write $a(i, j)$ for the expected task payoff of design i against j , $m(i, j)$ for its expected number of unsafe actions, and $u(i, j) = \mathbb{E}[n^U/T]$ for its expected unsafe frequency. Tables 1 and 2 report a and m . They are the only empirical input to every closed form below, so we place them in the main text; the twelve invasion thresholds they generate are collected in Appendix A. Two features of m carry most of the results: it vanishes identically on $\{\text{AS}, \text{CS}\}$, and $m(\text{CS}, \text{CAS}) = 4.222 > 0$ – a conditionally safe design is recorded as unsafe when it retaliates.

Table 1: Task payoffs $a(i, j)$ of the reduced race game at $p_r^{\max} = 0.6$, computed exactly over the horizon distribution. Rows index the focal design.

	AS	AU	CS	CAS
AS	59.000	5.400	59.000	8.600
AU	48.640	27.200	47.360	27.200
CS	59.000	16.600	59.000	26.644
CAS	101.765	27.200	61.631	27.200

Table 2: Expected number of Unsafe actions $m(i, j)$ of the focal design at $p_r^{\max} = 0.6$.

	AS	AU	CS	CAS
AS	0.000	0.000	0.000	0.000
AU	9.000	9.000	9.000	9.000
CS	0.000	8.000	0.000	4.222
CAS	1.000	9.000	4.778	9.000

2.2 Deployment layer: three functionals and the wedge

Each unsafe action imposes an external harm $h \geq 0$ on parties outside the race. Let $\lambda \in [0, 1]$ be the fraction of that harm borne by the principal who deployed the design – the liability pass-through. That single parameter splits the interaction into three functionals:

$$\text{task payoff: } a(i, j), \quad (3)$$

$$\text{selection functional: } \pi_P(i, j) = a(i, j) - \lambda h m(i, j), \quad (4)$$

$$\text{social functional: } \pi_S(i, j) = a(i, j) - h m(i, j). \quad (5)$$

Only π_P enters the dynamics. The social functional (5) is an accounting device: since each agent’s harm is attributed to itself, the population mean of π_S equals mean task payoff minus total external harm.

Definition 2 (Selection wedge). The selection wedge is $\Delta(i, j) = \pi_P(i, j) - \pi_S(i, j) = (1 - \lambda)h m(i, j)$.

Figure 1 summarises the construction, and Figure 2 shows the three functionals and the wedge for the baseline race. The wedge is the entire formal content of the separation between the layer at which safety is engineered and the layer at which designs are selected.

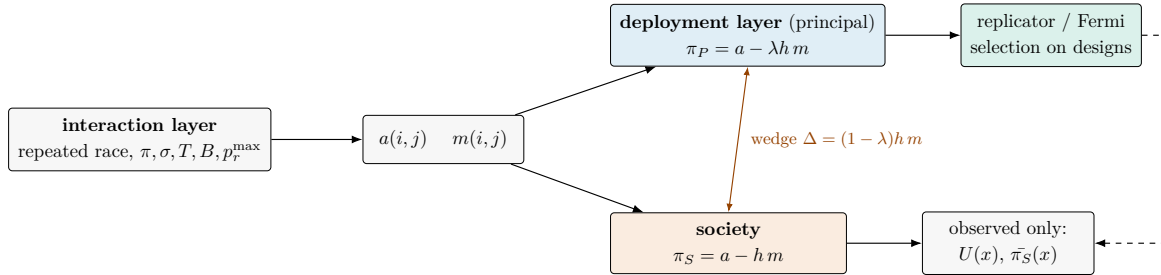


Figure 1: Deployment-layer selection. The interaction layer produces a task payoff a and an externality intensity m . The principal selects designs on π_P , which internalises a fraction λ of the harm; society is scored by π_S , which internalises all of it. Only the upper path feeds back into the population dynamics, so the social functional and the unsafe frequency $U(x)$ are observables of a system that does not optimise them. Their difference is the selection wedge Δ .

2.3 Dynamics

Let $x \in \Delta_n$ be the frequency distribution over the n available designs. In an infinite population the deployment dynamics is the replicator equation of the selection functional,

$$\dot{x}_i = x_i \left[(\pi_P x)_i - x^\top \pi_P x \right], \quad i = 1, \dots, n. \quad (6)$$

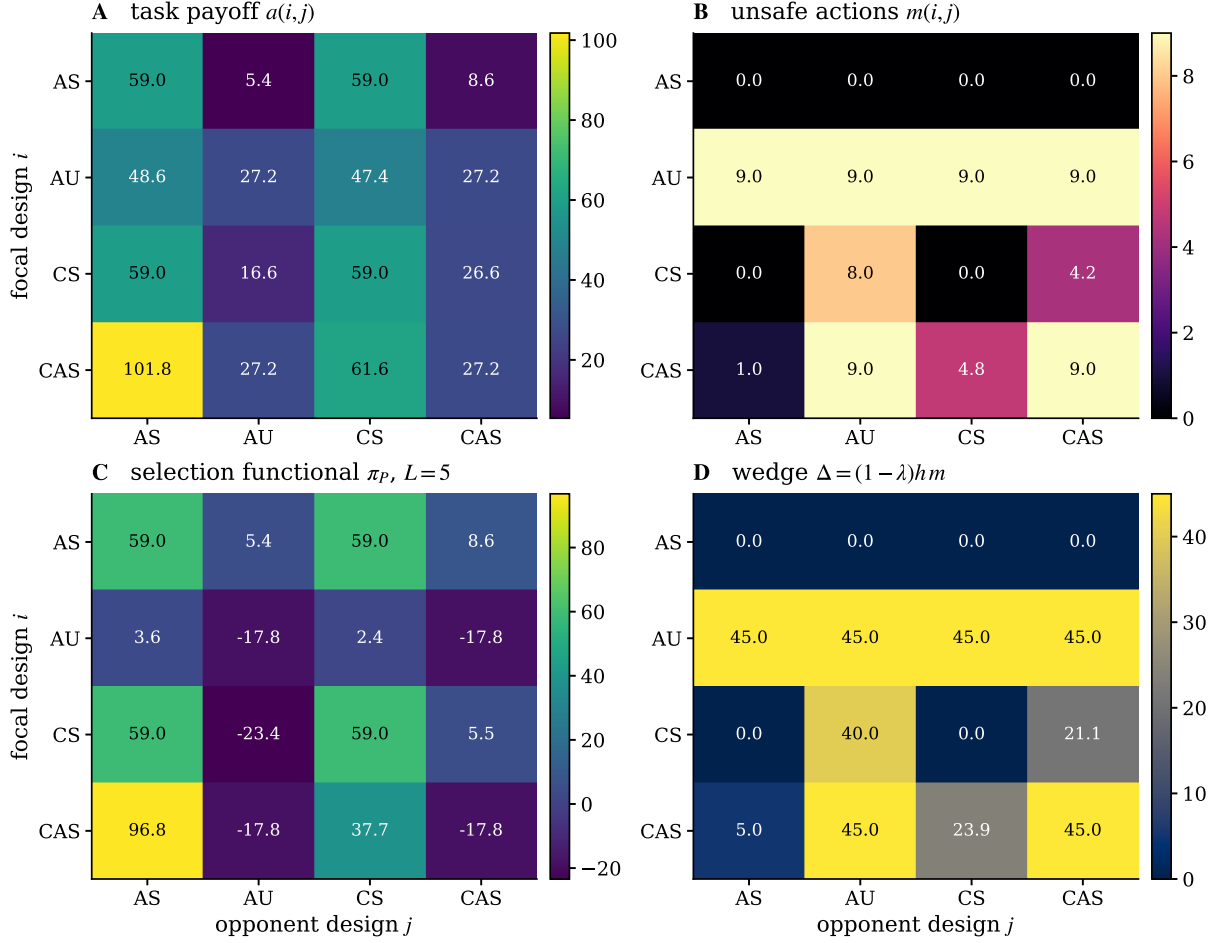


Figure 2: The three functionals of the baseline race ($B = 100$, $p_r^{\max} = 0.6$), rows indexed by the focal design. **(A)** Task payoff $a(i, j)$. **(B)** Expected number of unsafe actions $m(i, j)$. **(C)** Selection functional $\pi_P = a - Lm$ at effective liability $L = 5$. **(D)** The wedge $\Delta = (1 - \lambda)hm$ at $\lambda = 0.5$, $h = 10$; it varies down the columns, which by Lemma 4 is exactly why it deforms the dynamics.

In a finite population of size Z we use the pairwise-comparison process with Fermi imitation [36]: a design B adopts A with probability $(1 + \exp[-\beta(f_A - f_B)])^{-1}$, where β is the intensity of selection and f is computed from π_P , with mutation probability μ . We report both the full chain and its small-mutation limit [37]. All computations use EGTtools [38].

The primary observable is the population unsafe frequency

$$U(x) = \sum_{i,j} x_i x_j u(i, j), \quad (7)$$

which requires no welfare weights and is therefore independent of the uncalibrated harm parameter.

2.4 Numerical methods

Replicator trajectories are integrated with an adaptive LSODA solver at relative tolerance 10^{-10} and absolute tolerance 10^{-12} , to $t = 2000$, which is two orders of magnitude beyond the slowest relaxation we observe. Basin averages are taken over initial conditions drawn uniformly from the simplex, with 110 draws per parameter value in Figure 5 and 100 in Figure 3, from a fixed seed; doubling the number of draws changes the reported averages by less than 0.01. Where

pools of different size are compared (Figure 3) initial conditions are drawn on the common $\{\text{AS}, \text{CS}, \text{CAS}\}$ simplex and the extra design is seeded at a fixed share, so that the comparison is between one ecosystem with and without that design rather than between two different sampling measures; the reported curves use a 5% seed, and Section 3.6 reports the sensitivity to that choice. Finite-population stationary distributions are obtained by solving the singular linear system for the stationary vector of the sparse transition matrix, since densifying it is infeasible at the state-space sizes used here. The best-response dynamics of Section 3.8 has a discontinuous field, which defeats adaptive step-size control; it is integrated with a fixed step of 10^{-2} , which is exact up to the step size because the flow is a linear contraction on each best-response region. The noisy race of Section 3.7 is evaluated by exact propagation of the joint chain rather than by simulation, with the horizon truncated at 80 rounds, leaving a residual mass of 5×10^{-8} . Every number quoted in the text is regenerated by the released code.

3 Results

The results below are of three kinds, and confusing them would misread the paper. Some are theorems about arbitrary symmetric matrix games and carry no numbers at all. Some are exact statements about the race of Section 2 whose *form* is general but whose constants come from one parameterisation. And some are numerical findings that hold on the grids we swept and nowhere by proof. Table 3 sorts every claim of this section into those three classes, so that a reader can see at a glance which conclusions would survive a different race and which would have to be recomputed.

3.1 Reduction, inertness and invasion thresholds

This section assembles the analytical scaffolding. The first two statements are immediate or classical and are recalled rather than claimed; the closed forms in Lemma 7 and the general criterion in Lemma 16 do the work later.

Proposition 3 (Reduction to the effective liability). *The dynamics (6) and the Fermi process depend on (λ, h) only through the product $L = \lambda h$.*

Proof. Both dynamics depend on π_P only, and by (4) $\pi_P = a - Lm$. □

We call L the *effective liability*, measured in payoff units per unsafe action. It is the only policy variable in what follows; a regulator can move it by raising the pass-through λ at fixed harm, and the same outcome obtains for any (λ, h) on the hyperbola $\lambda h = L$. Because L inherits the scale of the race, we also report L/B .

Lemma 4 (Column-constant invariance; classical). *Let $A' = A + \mathbf{1}c^\top$ for some $c \in \mathbb{R}^n$. Then A and A' induce the same replicator field on Δ_n and have the same symmetric Nash set.*

Proof. $(A'x)_i = (Ax)_i + c^\top x$ for every i , so all fitnesses shift by the same state-dependent constant and the differences $(A'x)_i - x^\top A'x$ are unchanged; likewise every pairwise payoff comparison at a fixed opponent distribution is unchanged [3]. □

Definition 5. A wedge Δ is *strategically inert* if $\Delta = \mathbf{1}c^\top$, that is, if $\Delta(i, j)$ does not depend on the focal design i .

Table 3: Scope of the claims. “General” means a statement about symmetric matrix games or about the selection functional as such; “exact, this race” means an exact consequence of the matchup tables of Section 2, so that the form is general but the constants are not; “numerical” means a finding established by sweeping a grid. The baseline parameterisation throughout is $p_{\max} = 0.6$, $B = 100$ and the stage payoffs of Table 1; Section 3.6 varies the first two.

Claim	Class	Reading
Prop. 3, Lem. 4, Prop. 6	general	properties of any charge levied on a matrix game
Lem. 7, Prop. 8	general	affine invasion conditions; thresholds independent of the imitative dynamics
Lem. 16	general	mixture-free invasion at any twin face
Prop. 20	general	which measurement errors a selection functional absorbs
Thm. 10, Cor. 11	exact, this race	holds for the four reduced designs; fails once the pool contains a longer punisher (Section 3.7)
Prop. 13, Thm. 14	exact, this race	neutrality is structural; the ratio 77.7 is not
Thm. 17, Cor. 18	exact, this race	the window (4.27, 42.77) is this parameterisation; a window exists at all 24 grid points of Section 3.6
Prop. 12	exact, this race	the two-valued margin is a property of these four designs
Prop. 19	exact, this race	neutrality breaks at first order in any perturbation; the coefficients 379.8 and 8 are this race’s
Prop. 21, eq. (20)	exact, this race	closed forms in q and r ; the constants are this race’s
Thm. 22	general	given the stated hypotheses on the erosion channel
Sections 3.7–3.9, numerical parts	numerical	valley depths, basin volumes, escape times, curves

Proposition 6 (Inertness only at full pass-through). *In the reduced race game, $\Delta = (1 - \lambda)hm$ is strategically inert if and only if $\lambda = 1$ or $h = 0$. The deployment dynamics is therefore orbit-equivalent to the dynamics of the social functional exactly at full liability pass-through.*

Proof. By Lemma 4 inertness holds iff $m(i, j)$ is independent of i or the prefactor vanishes. Table 2 gives $m(\text{AS}, \text{AS}) = 0 \neq 9 = m(\text{AU}, \text{AS})$, so m depends on the focal design and inertness requires $(1 - \lambda)h = 0$. $\square$

Proposition 6 has a practical reading: there is no cheap partial liability that reproduces the socially correct dynamics. Any $\lambda < 1$ deforms the vector field, and the deformation is proportional to $1 - \lambda$. It does not follow that the deformation is harmful in proportion to its size; Section 3.4 shows the opposite over a wide range of L . The proposition also identifies, by

contraposition, the one liability rule that would be dynamically harmless: a rule whose charge depends on the counterparty's behaviour rather than on one's own is column-constant and, by Lemma 4, leaves the replicator field untouched.

Lemma 7 (Affine invasion conditions). *A rare design i in a resident population of j has positive growth rate under (6) if and only if*

$$\delta a - L \delta m > 0, \quad \delta a := a(i, j) - a(j, j), \quad \delta m := m(i, j) - m(j, j). \quad (8)$$

If $\delta m \neq 0$ the critical effective liability is $L^ = \delta a / \delta m$; the invader is favoured for $L < L^*$ when $\delta m > 0$ and for $L > L^*$ when $\delta m < 0$. If $\delta m = 0$ the outcome does not depend on L .*

Proof. The growth rate of a rare i is $\pi_P(i, j) - \pi_P(j, j)$, which by (4) equals $\delta a - L \delta m$. The case distinction follows by solving a linear inequality. $\square$

Nothing in Lemma 7 is specific to the replicator equation. The next remark records what that buys, because it determines how far the results below depend on the imitation rule.

Proposition 8 (The thresholds are properties of π_P). *Call a dynamics imitative and payoff-monotone if $\dot{x}_i = x_i g_i(x)$ with $\sum_i x_i g_i(x) = 0$ and $g_i(x) \geq g_j(x) \iff (\pi_P x)_i \geq (\pi_P x)_j$; the replicator equation, pairwise-proportional imitation and imitation of the better all qualify. Then:*

- (i) *for any such dynamics, a design weakly dominated under π_P never gains share on the design that dominates it, a face on which π_P is constant is a set of rest points, and a rare third design at a rest point of a two-design face grows exactly under the affine condition (8);*
- (ii) *for the best-response dynamics $\dot{x} = \text{BR}(x) - x$, whose rest points are the symmetric Nash equilibria, the interior rest point of the $\{\text{AS}, \text{CAS}\}$ face is a Nash equilibrium of the three-design game if and only if $L \geq L^\dagger$, so the bistable window opens at the same threshold.*

Every threshold in this paper is therefore the value of L at which a fitness difference under π_P changes sign, and not an artefact of the imitation rule.

Proof. For (i): weak dominance gives $(\pi_P x)_i \geq (\pi_P x)_j$ at every x , hence $g_i \geq g_j$ and x_i/x_j non-decreasing. On a face where π_P is constant all g_i are equal, and $\sum_i x_i g_i = 0$ forces them to vanish. At a rest point of a two-design face the residents have equal fitness, so the entrant's g has the sign of its fitness difference from theirs, which is (8). For (ii): p^* is a symmetric Nash equilibrium of the three-design game iff no design does strictly better against it than the residents; by Lemma 16 the only candidate is CS, whose advantage is $p^*(\pi_P(\text{CS}, \text{CAS}) - \pi_P(\text{AS}, \text{CAS}))$, positive exactly for $L < L^\dagger$ by Theorem 17. $\square$

The restriction to imitative dynamics in (i) is not cosmetic. Under the logit and best-response dynamics a design at vanishing frequency is replenished at a rate that does not vanish with its frequency, so the neutral safe face is not a rest set: both contract it to its barycentre. Neutrality of the face is a property of the payoffs, but the drift story of Section 3.3 requires imitation. Aspiration-driven learning, in which agents compare their payoff to a reference level rather than to each other [39], satisfies neither hypothesis. Section 3.8 therefore tests all of them numerically [40].

Applying Lemma 7 to every ordered pair gives Table 10. Four of the twelve thresholds organise the rest of the paper:

$$\begin{array}{ccccc}
\underbrace{L_{\text{CS} \rightarrow \text{CAS}}^* = 0.116}_{\text{conditional safety enters}} & < & \underbrace{L_{\text{CAS} \rightarrow \text{CS}}^* = 0.551}_{\text{conditional safety protected}} & & \\
< & \underbrace{L^\dagger = 4.274}_{\text{retaliation becomes too costly}} & < & \underbrace{L_{\text{CAS} \rightarrow \text{AS}}^* = 42.765}_{\text{unconditional safety protected}} & .
\end{array} \tag{9}$$

The spread of (9) over a factor of 368 is the quantitative signature of the results below. In units of the prize the outer thresholds are $L_{\text{CAS} \rightarrow \text{CS}}^*/B = 0.0055$ and $L_{\text{CAS} \rightarrow \text{AS}}^*/B = 0.428$.

3.2 Solo evaluation cannot move the equilibrium

Pre-deployment safety evaluation probes a model in isolation and admits it when its behaviour is acceptable. We formalise this as a filter on the design pool and show that it is dynamically inert.

Definition 9 (Solo-evaluation filter). The solo score of design i is $s(i) = u(i, \text{AS})$, its unsafe frequency against a fully safe environment. The filter of strictness ε admits $\Theta_\varepsilon = \{i : s(i) \leq \varepsilon\}$.

In the reduced game, $s(\text{AS}) = s(\text{CS}) = 0$, $s(\text{AU}) = 1$, and CAS defects exactly once before its opponent's safe play locks it into safe play, so

$$s(\text{CAS}) = \mathbb{E}\left[\frac{1}{T}\right] = \sum_{g \geq 0} \frac{0.2 \cdot 0.8^g}{5 + g} = 0.132. \tag{10}$$

Any filter with $\varepsilon \in [0.132, 1)$ therefore admits $\{\text{AS}, \text{CS}, \text{CAS}\}$ and rejects only AU. The next result says that the rejection is dynamically empty.

Theorem 10 (Weak dominance of CAS over AU). *For every $L \geq 0$ and every opponent j , $\pi_P(\text{CAS}, j) \geq \pi_P(\text{AU}, j)$, with strict inequality for $j \in \{\text{AS}, \text{CS}\}$.*

Proof. From Tables 1 and 2,

$$\pi_P(\text{CAS}, j) - \pi_P(\text{AU}, j) = \begin{cases} 53.125 + 8L, & j = \text{AS}, \\ 14.271 + 4.222L, & j = \text{CS}, \\ 0, & j \in \{\text{AU}, \text{CAS}\}, \end{cases} \tag{11}$$

each of which is non-negative for $L \geq 0$ and strictly positive in the first two cases. Against AU and CAS the two designs generate identical action paths from round two onwards and identical unsafe counts, so the difference vanishes identically. $\square$

The mechanism behind Theorem 10 is transparent: AU and CAS behave the same way against an unsafe environment, but CAS stops defecting against a safe one, which earns it more and costs it less. Unconditional aggression is dominated by conditional aggression at every liability level.

Corollary 11 (Filter futility). *Let $\Theta = \{\text{AS}, \text{AU}, \text{CS}, \text{CAS}\}$ and $\varepsilon \in [0.132, 1)$, so that $\Theta_\varepsilon = \{\text{AS}, \text{CS}, \text{CAS}\}$. Then, for every $L \geq 0$:*

- (i) along every interior solution of (6) the ratio $x_{\text{AU}}/x_{\text{CAS}}$ is non-increasing, and strictly decreasing whenever $x_{\text{AS}} + x_{\text{CS}} > 0$;
- (ii) any attractor of Θ that the filter removes carries the maximal unsafe frequency $U = 1$, so the filter cannot delete an attractor with lower harm than one it leaves behind;
- (iii) under a start measure common to both pools, the basin-averaged long-run unsafe frequencies of Θ and Θ_ε agree to within 1.2×10^{-3} across the whole L -axis.

Proof. For (i), (11) gives $\frac{d}{dt} \log(x_{\text{AU}}/x_{\text{CAS}}) = \pi_P(\text{AU}, x) - \pi_P(\text{CAS}, x) = -[(53.125 + 8L)x_{\text{AS}} + (14.271 + 4.222L)x_{\text{CS}}] \leq 0$. For (ii), by (i) an ω -limit with $x_{\text{AU}} > 0$ must satisfy $x_{\text{AS}} = x_{\text{CS}} = 0$ and hence lie in the $\{\text{AU}, \text{CAS}\}$ edge, on which the two designs are payoff-identical, every point is a rest point, and $u \equiv 1$. Part (iii) is numerical and reported in Figure 3A. $\square$

Part (ii) is weaker than one might hope, and the weakness is itself informative. Filtering AU does delete an attractor: for $L < L_{\text{AS} \rightarrow \text{AU}}^* = 2.422$ an AU-dominated population is neutrally stable, and removing AU removes it. But that attractor has $U = 1$, exactly the unsafe frequency of the CAS vertex that the filter leaves in place whenever $L < L_{\text{CS} \rightarrow \text{CAS}}^* = 0.116$. The filter changes which design occupies the all-unsafe state, not whether the state exists, and the state is reachable only from populations already dominated by AU.

Figure 3 makes the comparison on a common start measure. On that comparison the curves coincide over the whole L -axis. Panel B shows why. A solo evaluation sees $s(\text{CAS}) = 0.132$ and cannot distinguish CAS from a safe design at any threshold that admits reciprocation at all, while CAS in self-play produces $u = 1$ – the same equilibrium harm as the design the filter did remove.

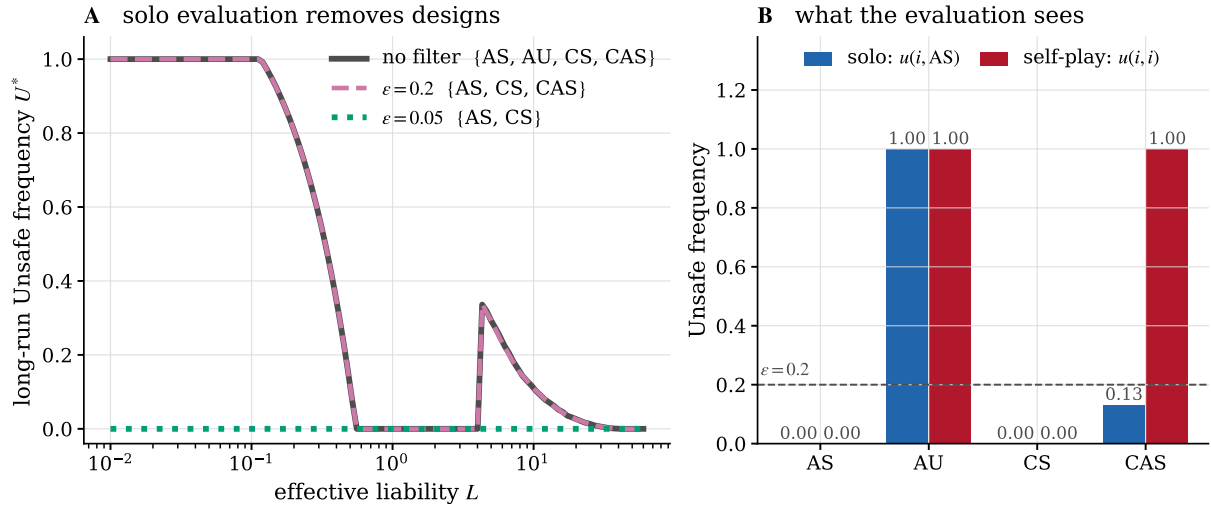


Figure 3: Solo evaluation filters are dynamically empty. **(A)** Long-run unsafe frequency of the unfiltered pool $\{\text{AS}, \text{AU}, \text{CS}, \text{CAS}\}$ and of the pools admitted at $\varepsilon = 0.2$ and $\varepsilon = 0.05$. Initial conditions are drawn on the $\{\text{AS}, \text{CS}, \text{CAS}\}$ simplex for every pool, with AU seeded at a fixed 5% where it is present, so that the three curves describe the same ecosystem. The first two coincide everywhere, as Corollary 11(iii) requires. **(B)** What the evaluation sees. Blue: unsafe frequency against a safe environment, the quantity the filter measures; the AS and CS bars are zero. Red: unsafe frequency in self-play, the quantity that determines equilibrium harm. The two orderings disagree precisely for CAS.

A strict filter with $\varepsilon < 0.132$ does remove CAS, leaving the pool $\{\text{AS}, \text{CS}\}$ on which $U^* = 0$ for every L . But the margin it works on is one defection in the first round: the whole difference

between the two scores in (10) is that CAS opens unsafely and is then locked into safe play. Section 3.7 shows what an execution error rate of a few per cent does to that margin.

What a multi-agent evaluation would buy. The futility of the solo filter is a statement about the probe, not about evaluation as such. Replace the single safe probe by a probe set P and score a design by its worst behaviour, $s_P(i) = \max_{j \in P} u(i, j)$. A filter of strictness ε then admits $\{i : s_P(i) \leq \varepsilon\}$, and it separates the safe pool from the unsafe one exactly on the band of strictnesses

$$\left[\max\{s_P(\text{AS}), s_P(\text{CS})\}, \min\{s_P(\text{AU}), s_P(\text{CAS})\} \right), \quad (12)$$

whose width is the margin the evaluation has to work with.

Proposition 12 (The probe must reciprocate). *Over all fifteen non-empty probe sets, the margin of (12) takes exactly two values. It equals 0.539 when P contains a conditional design and does not contain AU, and 0.132 otherwise. In particular*

$$\begin{aligned} P = \{\text{AS}\} : \quad s_P &= (0, 0, 1, 0.132), \quad \text{margin } 0.132, \\ P = \{\text{CS}\} : \quad s_P &= (0, 0, 1, 0.539), \quad \text{margin } 0.539, \\ P = \{\text{AS}, \text{AU}\} : \quad s_P &= (0, 0.868, 1, 1), \quad \text{margin } 0.132, \end{aligned} \quad (13)$$

with entries ordered (AS, CS, AU, CAS). Every strictness inside the band admits exactly {AS, CS}, which has $U^* = 0$ at every L .

Proof. $s_P(\text{AS}) = 0$ and $s_P(\text{AU}) = 1$ for every P , so the margin is $\min\{1, s_P(\text{CAS})\} - s_P(\text{CS})$. From Tables 1–2 the relevant unsafe frequencies are $u(\text{CS}, \cdot) = (0, 0.868, 0, 0.461)$ and $u(\text{CAS}, \cdot) = (0.132, 1, 0.539, 1)$ over opponents (AS, AU, CS, CAS). If $\text{AU} \in P$ then $s_P(\text{CS}) = 0.868$ and $s_P(\text{CAS}) = 1$, giving 0.132. Otherwise $s_P(\text{CAS})$ is 0.132 when $P = \{\text{AS}\}$ and at least 0.539 as soon as P contains CS or CAS, while $s_P(\text{CS})$ is 0 in the first case and at most 0.461 in the second; the two cases give 0.132 and 0.539. The full enumeration is reported with the code. The admitted pool has $m \equiv 0$ by Proposition 13, hence $U^* = 0$. $\square$

Proposition 12 is the constructive counterpart of Corollary 11, and it is more specific than “evaluate in a multi-agent setting”. What quadruples the margin is that the probe *reacts*: a conditionally safe counterpart answers the entrant’s opening defection and thereby elicits the condition that defines CAS, whereas an unconditionally safe counterpart absorbs it. What destroys the gain is hostility: adding an always-unsafe probe drives the score of conditional safety to 0.868, because it retaliates throughout, and the filter can no longer tell the guard from the aggressor. The strongest evaluation in this model is therefore neither the gentlest nor the harshest environment but the reciprocating one, and it is available at the cost of a single extra probe.

Two caveats remain, and both are structural rather than numerical. First, the admitted pool is exactly the neutral safe face of Section 3.3, so the protection lasts only as long as no CAS-like design re-enters, and Theorem 14 bounds how much liability is needed when one does; gating must be continuous rather than one-off. Second, the wide band tolerates a design that is unsafe in 46% of rounds against an aggressor, which is exactly the retaliation that liability taxes. An evaluation rule that scores retaliation as harm and a liability rule that charges it work against each other, and the model says they should not: the retaliation is what makes the safe pool cheap to protect.

3.3 A neutral safe face with a drifting barrier

Proposition 13 (Payoff twins). *Restricted to $\{AS, CS\}$, the selection functional is constant: $\pi_P(i, j) = a(AS, AS)$ for all $i, j \in \{AS, CS\}$ and all L . The AS – CS edge is therefore a set of rest points of (6), and in the finite-population process it evolves by neutral drift.*

Proof. A conditionally safe design plays safe in round one and then copies a safe opponent, so all four matchups within $\{AS, CS\}$ realise the all-safe path. Hence $a \equiv 59$, $m \equiv 0$ and $\pi_P \equiv 59$ independently of L . $\square$

Neutrality does not make the composition irrelevant, because AS and CS differ against an invader.

Theorem 14 (Composition-dependent invasion barrier). *Let the resident population be the mixture $(1 - x)AS + xCS$ on the safe face, and write $\delta a_k = a(CAS, k) - a(AS, AS)$ and $\delta m_k = m(CAS, k)$ for $k \in \{AS, CS\}$. A rare CAS invader has positive growth rate if and only if $L < L^*(x)$, where*

$$L^*(x) = \frac{(1 - x)\delta a_{AS} + x\delta a_{CS}}{(1 - x)\delta m_{AS} + x\delta m_{CS}}, \quad \begin{array}{ll} \delta a_{AS} = 42.765, & \delta m_{AS} = 1, \\ \delta a_{CS} = 2.631, & \delta m_{CS} = 4.778. \end{array} \quad (14)$$

L^* is strictly decreasing on $[0, 1]$, with $L^*(0) = 42.765$ and $L^*(1) = 0.551$, a ratio of 77.7.

Proof. By Proposition 13 the resident fitness is $a(AS, AS)$ for every x , so the invader's growth rate is $(1 - x)(\delta a_{AS} - L\delta m_{AS}) + x(\delta a_{CS} - L\delta m_{CS})$, which is positive exactly when $L < L^*(x)$. Differentiating the ratio of the two affine functions, the sign of $\frac{d}{dx}L^*$ is that of $\delta a_{CS}\delta m_{AS} - \delta a_{AS}\delta m_{CS} = 2.631 - 204.34 < 0$. $\square$

Inverting (14) gives the critical share of conditional safety $x^*(L)$ that a given liability level requires: $x^*(1) = 0.951$, $x^*(2) = 0.855$, $x^*(5) = 0.640$, $x^*(20) = 0.197$. A population that is 90% unconditionally safe is invadable at any $L < 28.1$, whereas the same population with the shares reversed is already protected at $L = 1.6$.

The combination of Proposition 13 and Theorem 14 is the substantive point. The variable that determines whether a safe population can be invaded carries no selection gradient of its own, so it is free to drift. Figure 4C shows the stationary distribution of the neutral edge in a finite population: for small mutation the mass concentrates at the two vertices, so the population spends long stretches near pure AS , which is the maximally vulnerable composition. Protection obtained by mandating unconditional safety is therefore not durable – not because the mandate is violated, but because the direction it protects against is selectively invisible.

Exact twinning is of course a knife-edge, and the reader is entitled to ask how much of this survives an approximate one. Proposition 19 answers that: neutrality is lost at first order in any perturbation, the face is then crossed in a time proportional to the inverse of the perturbation, and under execution noise the induced gradient points at the vertex that is cheap to protect. This section is therefore the worst case rather than the typical one, which is the right way to read it.

3.4 Bistability and the liability valley

The results so far might suggest that harm decreases with liability. It does not. This section identifies a window of intermediate liability on which the system is bistable and the basin-averaged

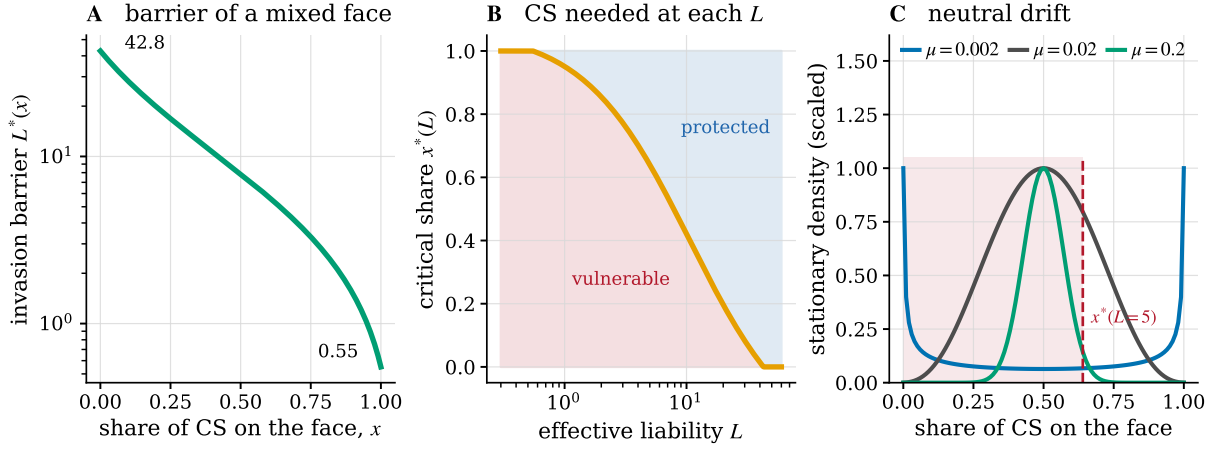


Figure 4: The safe face is neutral but not uniform. **(A)** Invasion barrier $L^*(x)$ of Theorem 14, log scale. **(B)** The critical share of conditional safety $x^*(L)$ separating vulnerable from protected compositions. **(C)** Stationary density of the neutral AS–CS edge in a finite population ($Z = 100$, $\beta = 0.05$) for three mutation rates, with the vulnerability threshold at $L = 5$ marked. Small mutation concentrates the mass at the vertices, half of it on the vulnerable side.

harm jumps upward.

Proposition 15 (Interior rest point of the unsafe face). *On the $\{AS, CAS\}$ face the replicator field has an interior rest point whenever $\alpha := \pi_P(CAS, AS) - \pi_P(AS, AS)$ and $\gamma := \pi_P(AS, CAS) - \pi_P(CAS, CAS)$ share a sign, located at $p^* = \alpha/(\alpha + \gamma)$ in the CAS direction, and it attracts the face when both are positive. With the values of Table 1,*

$$p^*(L) = \frac{42.765 - L}{24.165 + 8L}, \quad L \in (L_{AS \rightarrow CAS}^*, L_{CAS \rightarrow AS}^*) = (2.067, 42.765). \quad (15)$$

Proof. Direct computation of the two-strategy replicator rest point, followed by substitution of $\alpha = 42.765 - L$ and $\gamma = 9L - 18.6$. $\square$

The next statement is the mathematical core of the section, and it is not specific to the race game.

Lemma 16 (Mixture-free invasion at a twin face). *Let M be a symmetric matrix game, let i, j span a face with an interior rest point at frequency $p \in (0, 1)$ of j , and let k be a third strategy that is a payoff twin of i against i , that is $M_{ki} = M_{ii}$. Then the growth rate of a rare k at that rest point equals $p(M_{kj} - M_{ij})$. Its sign is therefore independent of p , and k invades the interior rest point if and only if $M_{kj} > M_{ij}$.*

Proof. At an interior rest point of the face the two residents have equal fitness $f = (1 - p)M_{ii} + pM_{ij}$. The growth rate of a rare k is $(1 - p)M_{ki} + pM_{kj} - f = (1 - p)(M_{ki} - M_{ii}) + p(M_{kj} - M_{ij})$, and the first term vanishes by hypothesis. $\square$

Lemma 16 says that whenever a candidate entrant is behaviourally indistinguishable from an incumbent against that incumbent, whether it can enter a mixed equilibrium involving the incumbent is decided by a single pairwise comparison, with no dependence on the equilibrium mixture. Applied to our game with $i = AS$, $j = CAS$, $k = CS$ – the hypothesis is exactly Proposition 13 – it yields the threshold that governs the valley.

Theorem 17 (Guard threshold). *CS invades the interior rest point of the $\{AS, CAS\}$ face if and only if*

$$L < L^\dagger := \frac{a(CS, CAS) - a(AS, CAS)}{m(CS, CAS) - m(AS, CAS)} = \frac{26.644 - 8.600}{4.222 - 0} = 4.274. \quad (16)$$

Proof. Apply Lemma 16 with $M = \pi_P$, $i = AS$, $j = CAS$, $k = CS$; the twin hypothesis $\pi_P(CS, AS) = \pi_P(AS, AS)$ is Proposition 13. The criterion $\pi_P(CS, CAS) > \pi_P(AS, CAS)$ is affine in L by (4), and solving it gives (16). $\square$

The interpretation of $L^\dagger$ is the crux of the paper. A conditionally safe design defends a population by retaliating: it matches unsafe play, which is what makes exploiting it unprofitable. Retaliation is itself unsafe behaviour, so liability taxes it. Above $L^\dagger$ the tax on retaliation exceeds the tax the retaliation imposes on the aggressor, the guard loses its advantage, and the unsafe coexistence becomes unavoidable.

Corollary 18 (Liability valley). *For $L \in (L^\dagger, L_{CAS \rightarrow AS}^*) = (4.274, 42.765)$ the dynamics on $\{AS, CS, CAS\}$ has two attractors: the neutral safe face with $U = 0$, and the interior rest point (15) with $U = p^*(1 - p^*)u(CAS, AS) + (p^*)^2 > 0$. At the lower edge the unsafe attractor carries $U = 0.465$; weighting the two attractors by the volume of their basins, the long-run unsafe frequency is discontinuous at $L^\dagger$, rising from 0 to 0.32 as L crosses it from below.*

Corollary 18 is the sharpest policy statement available from the model: over an interval spanning a factor of ten in L , more liability means more harm. A regulator moving L upward from a protected regime passes through a range in which the protection it was relying on disappears. It does not recover that protection until L exceeds 42.765, a level at which the external harm of a single unsafe development step is worth more than 42% of the prize for winning the race.

Figure 5 assembles the analytic branches and the two numerical dynamics. The agreement is exact where it should be: the basin-averaged replicator tracks the analytic CS–CAS branch on (0.116, 0.551) and the analytic AS–CAS branch inside the valley, lying slightly below it because a fraction of initial conditions still reaches the safe attractor. Figure 6 shows the corresponding phase portraits; the thick dashed edge visible in every panel is the neutral safe face of Proposition 13, drawn as a continuum of rest points.

3.5 Finite populations

The deterministic picture assumes an infinite population and no exploration. Figure 7 reports the stationary value of the observable (7) for the Fermi process over the (L, β) and (L, Z) planes; by Proposition 3 the horizontal axis is again the effective liability alone.

The two thresholds $L_{CAS \rightarrow CS}^*$ and $L_{CAS \rightarrow AS}^*$ remain visible as the boundaries of the transition region for all β and Z examined. Weak selection smooths the transition into a broad crossover, as expected: at $\beta \rightarrow 0$ imitation is nearly random and the stationary distribution approaches the uniform one over states, so U^* approaches the value implied by a uniform design mixture rather than by any equilibrium. Increasing Z sharpens the transition, consistent with the deterministic limit.

Stochasticity also supplies the transitions between the two attractors of Corollary 18 that the deterministic flow forbids, so bistability becomes a statement about time scales. Table 4

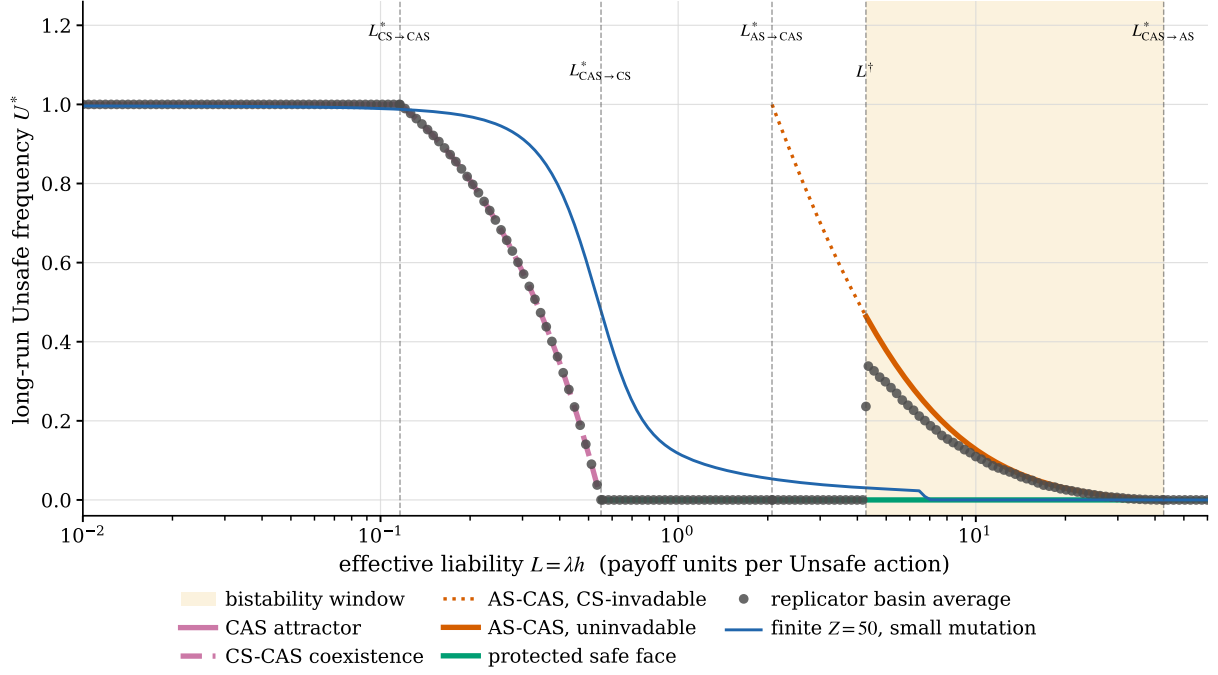


Figure 5: Bifurcation diagram in the effective liability. Thick curves are the analytic branches; markers are the basin-averaged replicator attractor over 110 uniform interior initial conditions; the thin line is the stationary unsafe frequency of the finite-population process ($Z = 50$, small-mutation limit). The shaded band is the bistability window ($L^\dagger, L_{\text{CAS} \rightarrow \text{AS}}^*$) of Corollary 18, on which raising liability raises expected harm.

quantifies it at $L = 10$, in the middle of the valley: starting from the unsafe rest point, the mean number of generations needed to reach a state free of CAS ranges from 65 at $Z = 30, \beta = 0.02$ to 3.9×10^{13} at $Z = 100, \beta = 0.2$, and the stationary mass on states in which CAS exceeds a quarter of the population rises from 0.29 to 0.997 over the same range. The valley is therefore not an artefact of the deterministic limit. It is a long-lived trapping region whose residence time grows steeply in both the population size and the intensity of selection, and it is only at weak selection in small populations that liability inside the window buys a protection that is merely slower rather than absent.

Table 4: Escape from the unsafe attractor inside the liability valley ($L = 10$, pool $\{\text{AS}, \text{CS}, \text{CAS}\}$, $\mu = 1/Z$). Mean exit time is the expected number of generations to reach a state free of CAS, starting from the interior rest point of Proposition 15. Unsafe mass is the stationary probability that CAS exceeds a quarter of the population.

Z	β	mean exit (generations)	stationary unsafe mass
30	0.02	6.5×10^1	0.291
30	0.20	2.1×10^4	0.942
50	0.05	5.9×10^2	0.636
80	0.05	5.4×10^3	0.768
100	0.05	2.5×10^4	0.833
100	0.20	3.9×10^{13}	0.997

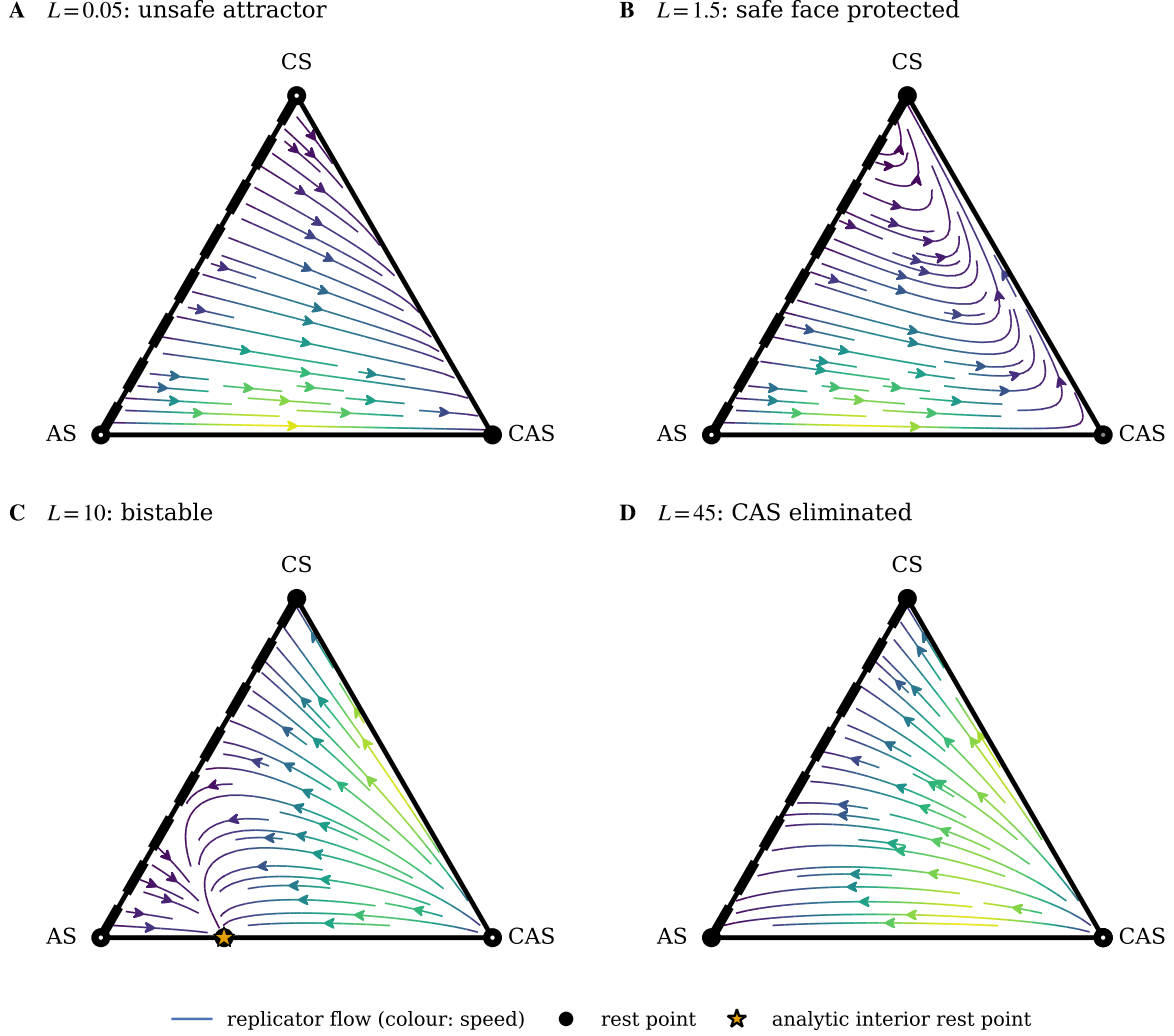


Figure 6: Replicator phase portraits on the $\{AS, CS, CAS\}$ simplex at four liability levels, drawn with the EGTtools simplex utilities [38]. Colour encodes the local speed of selection, scaled within each panel; the thick dashed AS–CS edge is the neutral face. The star marks the analytic interior rest point of Proposition 15. In the valley (C) the flow separates into two basins; at high liability (D) every trajectory reaches the neutral safe face, where it stops at a history-dependent composition.

3.6 Robustness to the constants of the race

Every threshold in (9) is a ratio of payoff differences and therefore inherits the scale of the prize B and the shape of the private risk $p_r^{\max}$. Figure 8 recomputes them over $B \in [10, 400]$ and $p_r^{\max} \in \{0.1, 0.3, 0.6, 0.9\}$. Three findings.

First, the liability that protects unconditional safety is a roughly constant fraction of the prize: $L_{CAS \rightarrow AS}^*/B$ lies between 0.38 and 0.61 across the whole grid, decreasing weakly in B and in $p_r^{\max}$. The requirement is therefore not an artefact of the particular value $B = 100$; it says that deterring a single unsafe step in a winner-take-all race costs around half the prize, whatever the prize is.

Second, the advantage of conditional over unconditional safety is robust and grows with the prize: at $p_r^{\max} = 0.6$ the ratio rises from 31.7 at $B = 10$ to 90.4 at $B = 400$, and over the whole grid it ranges from 4.4 to 90.4. At $p_r^{\max} = 0.9$ the comparison degenerates

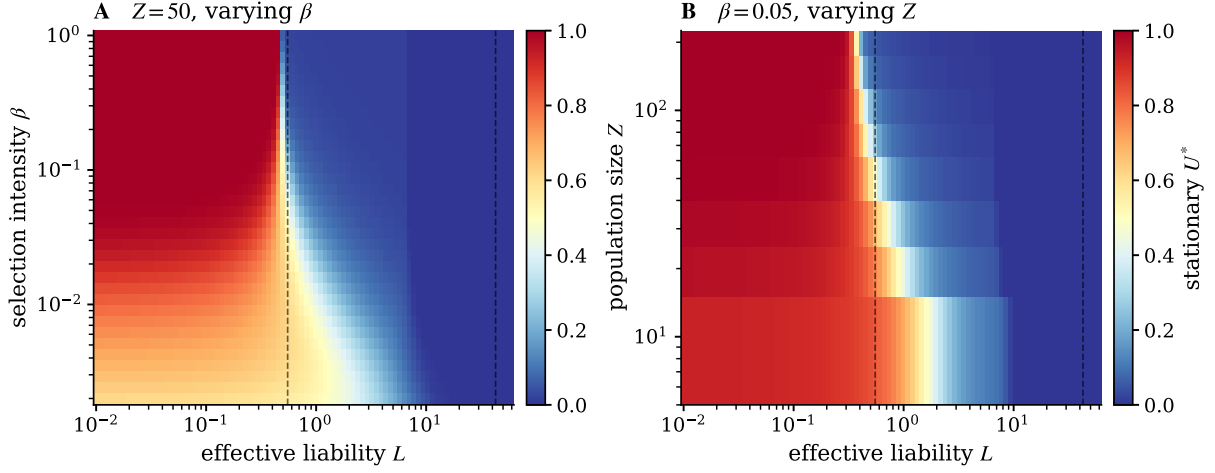


Figure 7: Stationary unsafe frequency of the finite-population pairwise comparison process on $\{AS, CS, CAS\}$ in the small-mutation limit. **(A)** Varying selection intensity at $Z = 50$. **(B)** Varying population size at $\beta = 0.05$. Dashed lines mark $L_{CAS \rightarrow CS}^*$ and $L_{CAS \rightarrow AS}^*$.

in the direction that strengthens the message: $L_{CAS \rightarrow CS}^*$ becomes negative, meaning that a conditionally safe population repels CAS with *no* liability at all, while unconditional safety still requires $L_{CAS \rightarrow AS}^* = 38.4$. That CS becomes uninvadable at high private risk also reproduces, from the selection layer, the treatment ordering reported for the interaction layer in the source study [18].

Third, the bistability window exists at every one of the 24 grid points, with multiplicative width between 4.19 and 12.79, widening in both B and $p_r^{\max}$. The liability valley is thus a structural feature of this class of races rather than a coincidence of the baseline parameters.

We also checked the sensitivity of Corollary 11(iii) to the share at which the filtered design is seeded. The gap between the filtered and unfiltered curves stays below 4.2×10^{-3} for seeds up to 20% at every L except $L = 0.3$, where it reaches 1.2×10^{-2} at a 20% seed and 0.12 at a 35% seed. The equivalence is a statement about an ecosystem into which the filtered design enters as a minority, which is the case of interest; it is not claimed for a population already dominated by that design.

3.7 Execution noise and richer designs

The four designs are deterministic, which is what makes the matchup tables exact and the safe face an exact twin face. Both properties are knife-edge, so we now break them. A design is promoted to a stochastic finite-state machine and every intended action is inverted with probability η before it is played and before the opponent observes it. The joint chain over the two internal states is propagated exactly against the horizon law, carrying the progress difference and the unsafe count as state and the accumulated stage payoff as a first moment, so the evaluation remains free of sampling noise; at $\eta = 0$ it reproduces Tables 1 and 2 to within 1.1×10^{-6} , which is the horizon truncation and nothing else. We also enlarge the pool with a generous guard GCS that forgives an observed unsafe action with probability 0.25, its aggressive counterpart GCAS, and a two-round punisher PUN2 that answers one observed unsafe action with two.

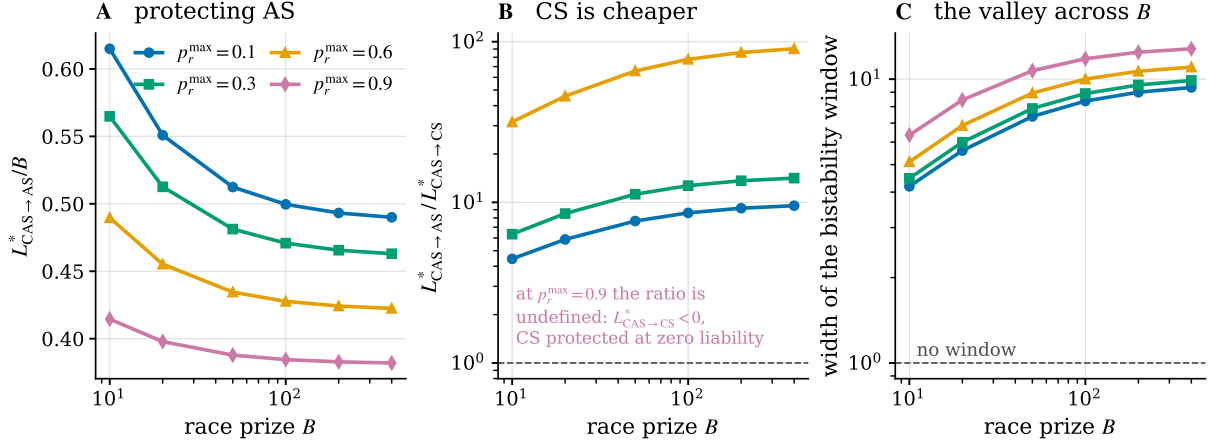


Figure 8: Thresholds across the constants of the race. **(A)** Liability protecting unconditional safety, in units of the prize. **(B)** Ratio $L_{CAS \rightarrow AS}^*/L_{CAS \rightarrow CS}^*$, the factor by which unconditional safety is cheaper to invade; the $p_r^{\max} = 0.9$ curve is absent because its denominator is negative, meaning conditional safety needs no liability at all. **(C)** Multiplicative width of the bistability window, which is greater than one at every grid point.

Noise breaks the safe face, in favour of conditional safety. With $\eta > 0$ the twins of Proposition 13 separate, because a conditionally safe design retaliates against its partner's trembles while an unconditionally safe one absorbs them. The separation has a sign. A rare CS strictly invades an AS resident for every L below 44.3 at $\eta = 0.01$, 41.4 at 0.02, 34.4 at 0.05 and 26.4 at 0.10, and the reverse holds above. Since the entire policy-relevant range lies below those values, noise converts the neutrality of Theorem 14 into a strict gradient pointing at the cheap-to-protect vertex. The drift argument of Section 3.3 is therefore the worst case, not the typical one – but the direction reverses at high liability, which is one more way in which a strong instrument selects for the fragile composition.

How fast the neutrality goes is a separate question from whether it goes, and it has a clean answer.

Proposition 19 (The rate at which twinning breaks). *Perturb the safe face so that the growth rate of a rare CS in an AS resident is $g = \delta a - L \delta m$ instead of zero. On the face the replicator equation reduces to $\dot{x} = x(1-x)g$, so the face is traversed from x_0 to $1-x_0$ in time $g^{-1} \ln((1-x_0)^2/x_0^2)$ and neutrality breaks at first order in the perturbation. Under execution noise the perturbation is*

$$\delta m = (\mathbb{E}[T] - 1)\eta + O(\eta^2) = 8.000\eta + O(\eta^2), \quad \delta a = 379.8\eta + O(\eta^2), \quad (17)$$

so $g = \eta(379.8 - 8L) + O(\eta^2)$ and conditional safety is favoured at every $L < 47.5$.

Proof. The face carries no other force, so the replicator field restricted to it is $x(1-x)$ times the fitness difference, which is g by Proposition 13 applied to the unperturbed part; integrating gives the time. For (17), a copier mirrors each tremble of its partner exactly once, in each of the $\mathbb{E}[T] - 1$ rounds that follow the first, whereas its own trembles are counted identically for both designs; δa is obtained from the exact evaluator, and both coefficients are confirmed to three digits by Richardson extrapolation on $\eta \in [10^{-4}, 10^{-3}]$. $\square$

The crossing time is thus $\Theta(1/\eta)$ with a computable constant: from a 5% minority to a 95% majority takes 15.5 time units at $\eta = 10^{-3}$ and 1.55 at $\eta = 10^{-2}$, at $L = 0$. In a finite population

under Fermi imitation, selection dominates drift on the face once $\beta Zg \gtrsim 1$, which at the baseline $\beta = 0.05$, $Z = 100$ and $L = 0$ means $\eta \gtrsim 5.3 \times 10^{-4}$. Exact twinning is therefore a knife-edge in the strict sense – a tremble rate of one in two thousand already replaces drift by selection – and the paper’s neutral face should be read as the limiting case of a slow directed drift towards the protected vertex rather than as a description of an ecosystem with no execution error at all.

The guard threshold is stable; the valley is not. Table 5 recomputes the thresholds. $L^\dagger$ moves by less than 13% over $\eta \in [0, 0.10]$ and the bistability window survives at every noise level examined, narrowing from a factor of 10.0 to 2.6 at $\eta = 0.20$. What does not survive is the non-monotonicity of *harm*. The reason is visible in the same table: the protected attractor stops being protected, since $u(\text{CS}, \text{CS})$ rises to 0.047, 0.088 and 0.186 at $\eta = 0.01, 0.02, 0.05$ – a single tremble triggers a retaliation cascade, so residual harm grows faster than the noise that causes it. The gap between the two attractors closes accordingly, and the largest increase of long-run harm along the L -axis falls from 0.299 at $\eta = 0$ to 0.149, 0.089 and 0.025, vanishing at $\eta \geq 0.075$ (Figure 9A). The liability valley is thus a feature of low-error ecosystems. That is not a reassurance: above the critical noise level liability becomes monotone again only because the state it was protecting has stopped being safe.

Table 5: Execution noise. Every quantity is recomputed exactly for designs that invert each intended action with probability η . $L^\dagger$ is the guard threshold, $L_{\text{CAS} \rightarrow \text{AS}}^*$ the liability protecting unconditional safety, and the window width their ratio; the barrier ratio is $L^*(0)/L^*(1)$ across the safe face. $u(\text{CS}, \text{CS})$ is the residual harm of a conditionally safe population, which is zero only at $\eta = 0$. The valley depth is the largest increase of the basin-averaged unsafe frequency along the L -axis; it vanishes at $\eta \geq 0.075$.

η	$L^\dagger$	$L_{\text{CAS} \rightarrow \text{AS}}^*$	width	barrier ratio	$u(\text{CS}, \text{CS})$	valley depth
0.000	4.27	42.77	10.01	77.7	0.000	0.299
0.010	4.21	39.20	9.31	63.7	0.047	0.149
0.020	4.19	36.16	8.62	55.4	0.088	0.089
0.050	4.30	29.30	6.82	44.4	0.186	0.025
0.075	4.50	25.32	5.63	42.5	0.247	0.000
0.100	4.75	22.34	4.70	44.0	0.294	0.000
0.150	5.35	18.23	3.41	56.5	0.361	0.000
0.200	6.02	15.59	2.59	89.8	0.405	0.000

The solo filter fails for a second reason. The strict filter of Section 3.2 separates CS from CAS by the single first-round defection, $s(\text{CAS}) - s(\text{CS}) = 0.132$. Under noise both scores rise and the gap is roughly preserved (0.127, 0.119 and 0.106 at $\eta = 0.02, 0.05, 0.10$), but its *location* moves: $s(\text{CS})$ reaches 0.130 at $\eta = 0.075$. A filter calibrated on the deterministic model therefore starts rejecting conditional safety once the error rate exceeds about 7.6% – and conditional safety is the design that Theorem 14 identifies as the cheapest to protect. The reciprocating probe of Proposition 12 degrades from a much higher base: its margin is 0.539 at $\eta = 0$ and still 0.449, 0.349 and 0.237 at $\eta = 0.02, 0.05$ and 0.10, so even at a ten per cent error rate it retains more margin than the solo probe has at no error at all.

The margin also fixes how much evaluation a filter has to buy, which is the form in which the difference between the two probes is easiest to act on. An evaluator that estimates each score

from n independent episodes separates two designs whose true scores differ by Δ at the 95% level once $n > (1.386/\Delta)^2$, using the Bernoulli bound $\text{sd} \leq 0.5$ on a frequency. The reciprocating probe needs $n \geq 7$ episodes at $\eta = 0$ and $n \geq 16$ at $\eta = 0.05$; the solo probe needs $n \geq 111$ and $n \geq 136$. Probing against a counterpart that reciprocates is therefore not only more reliable but roughly an order of magnitude cheaper for the same confidence, and the gap widens rather than narrows as execution becomes noisier.

Weak dominance is a property of the pool. Theorem 10 does not survive the enlarged pool, and the way it fails is instructive. Against PUN2, unconditional aggression earns 15.48 more than conditional aggression while committing one more unsafe action, because AU’s uninterrupted unsafe play wins the race outright whereas CAS, thrown out of phase by the two-round punishment, only ties. The dominance is restored for $L \geq 15.48$. The corollary that follows is weaker than the theorem but is the one that matters for policy: a filter that removes AU is dynamically empty *unless* the surviving pool contains a design that punishes for longer than it is punished. Building such designs is a more promising intervention than filtering, and it is an intervention on the design layer that the selection layer does not undo.

The valley survives enrichment, and moves. Adding each of the three designs to $\{\text{AS}, \text{CS}, \text{CAS}\}$, singly and together, leaves a discontinuous upward jump of long-run harm in L in every case. Its depth is 0.299 for the baseline pool, 0.344 with the generous guard, 0.163 with the harsh guard, 0.181 with the generous aggressor and 0.084 with all three; at $\eta = 0.02$ the corresponding depths are 0.089, 0.055, 0.033, 0.120 and 0.044. The location moves substantially – the edge sits at $L^\dagger = 4.27$ in the baseline pool and between 12 and 13 in the full pool – so a regulator cannot read the dangerous range off the reduced game. What is robust is that a dangerous range exists, and the harsh punisher shrinks it rather than removing it.

A design that never forgives. The extreme point of the retaliation axis is a grim trigger GRIM, which answers one observed unsafe action with unsafe play for the remainder of the interaction; it is the limit of PUN k as the punishment outlasts the horizon, and it is the strongest guard this action space admits, earning 31.88 against CAS where CS earns 26.64. Three things follow, and the first is the most useful. It moves no closed-form threshold at all: adding a design that is absent from the $\{\text{AS}, \text{CS}, \text{CAS}\}$ submatrix leaves $L^\dagger$ and $L_{\text{CAS} \rightarrow \text{AS}}^*$ exactly where they were, since both are computed from that submatrix alone. The thresholds of this paper are properties of a submatrix, and enlarging the pool moves basins rather than edges. Second, the basins it moves, it moves in the safe direction: the valley depth falls from 0.299 to 0.163 with GRIM in the pool and to 0.109 with a two-round punisher alongside it. Third, it is the design that execution noise damages most, because a single tremble is permanent: a population of grim triggers is unsafe in 7.6% of rounds at $\eta = 0.01$ and 28.9% at $\eta = 0.05$, against 4.7% and 18.6% for the copier. By $\eta = 0.05$ the valley in every enlarged pool has vanished, for the reason that is by now familiar: the state that liability was protecting has stopped being safe. Longer punishment is a better guard in a clean ecosystem and a worse citizen in a noisy one, and the error rate decides which effect wins.

3.8 Non-linear liability, other dynamics, and structure

Non-linear charges. Real liability is not linear in the number of incidents. Replace the charge by $L\varphi(m)$ for a strictly increasing φ with $\varphi(0) = 0$, normalised so that $\varphi(9) = 9$, which fixes the charge on a design that is unsafe throughout and makes L comparable across shapes. Lemma 7 survives verbatim with δm replaced by $\delta\varphi$, so every threshold is still a closed form, and the safe face is still an exact twin face because m vanishes identically on it and hence so does $\varphi(m)$. Table 6 reports five shapes. The valley exists in all of them, and the two ends of the range behave oppositely. A convex charge pushes the window from a factor of 10.0 to 42.3, because it barely charges the single unsafe action with which CAS exploits AS and so makes unconditional safety enormously expensive to protect. A concave or capped charge does the reverse: it charges the first incident heavily and saturates, which taxes initiation more than escalation, and it narrows the window to 4.9 and 7.1. A de minimis rule that charges nothing below two incidents never protects unconditional safety at any liability – its window is unbounded – since the exploiting action is exactly one incident. The shape of the charge is therefore a policy lever in its own right, and it moves in the direction the mechanism predicts: what shrinks the valley is a rule that is steep at the first incident and flat thereafter.

Table 6: Non-linear liability. The charge is $L\varphi(m)$ with φ increasing and normalised so that $\varphi(9) = 9$, which fixes the charge on a design that is unsafe throughout and makes L comparable across shapes. Every threshold remains a ratio of a payoff difference to a charge difference. A de minimis rule that does not charge a single incident never protects unconditional safety, so its window is unbounded.

charge shape	$L^\dagger$	$L_{\text{CAS} \rightarrow \text{AS}}^*$	width	valley depth
linear, $\varphi(m) = m$	4.27	42.77	10.01	0.255
convex, $\varphi \propto m^2$	9.11	384.89	42.25	0.112
concave, $\varphi \propto \sqrt{m}$	2.93	14.26	4.87	0.447
capped at $m = 3$	2.00	14.26	7.11	0.400
de minimis below $m = 2$	6.32	∞	∞	0.185

Other selection dynamics. Proposition 8 covers imitative dynamics exactly and locates the best-response bifurcation at the same threshold. Figure 9B confirms both and delimits them. Under the logit dynamics the valley has depth 0.417 at $\beta = 10$ and 0.171 at $\beta = 1$, and under best response with inertia 0.433, against 0.305 for the replicator; the edge sits at $L^\dagger$ in each case, the differences being in how much of the simplex each dynamics routes into the unsafe basin. Aspiration-driven learning [39], in which nobody compares payoffs with anybody, behaves differently: long-run harm is monotone in L but stops responding to it, settling at 0.237 for every $L \geq 4$ at an aspiration of 55. The failure mode is then not that the instrument backfires but that it loses traction, which is a distinct and equally unwelcome possibility.

Structured populations. The results above assume random matching. Under weak selection and rare mutation, the comparison of any two designs in a structured population is governed by a single structure coefficient $\sigma \geq 1$: k is favoured over i – in the sense that its fixation probability is the larger – if and only if $\sigma \pi_P(k, k) + \pi_P(k, i) > \pi_P(i, k) + \sigma \pi_P(i, i)$, where σ summarises the population structure and is 1 for a well-mixed population [41]. Two of our results can therefore

be carried over exactly. First, the safe face is neutral for *every* σ , because all four entries of π_P on $\{\text{AS}, \text{CS}\}$ are equal and the criterion reduces to an identity; neutrality is a property of the payoffs, not of the matching. Second, the criterion is affine in L for each σ and gives

$$\begin{aligned} \text{CAS favoured over AS} &\iff \sigma(31.8 + 9L) < 93.165 - L, \\ \text{CAS favoured over CS} &\iff \sigma(31.8 + 9L) < 34.99 - 0.556L, \end{aligned} \tag{18}$$

so the liability needed to keep conditional aggression out falls steeply with assortment: $L > 6.14$ suffices at $\sigma = 1$, $L > 3.14$ at $\sigma = 1.5$ and $L > 1.56$ at $\sigma = 2$, while for $\sigma \geq 2.93$ unconditional safety is protected with no liability at all. Against conditional safety the corresponding numbers are $L > 0.33$ at $\sigma = 1$ and no liability at all for $\sigma \geq 1.10$. Assortment and liability are substitutes here, and the substitution is strongly in favour of the conditional design. What the σ -criterion cannot settle is the valley, because it orders each pair of designs and a bistable pair has no order; for that we need a route that keeps the dynamics.

Assortative matching. The deterministic idealisation of structure supplies one. Under assortment at level r [42, 43] a design meets a copy of itself with probability r and a partner drawn at random otherwise, so its fitness is $r\pi_P(i, i) + (1 - r)\sum_j x_j \pi_P(i, j)$, which is the replicator dynamics of the transformed matrix

$$\tilde{\pi}(i, j) = r\pi_P(i, i) + (1 - r)\pi_P(i, j). \tag{19}$$

The transform is linear and acts on a and m separately, so $\tilde{\pi}$ is still affine in L and every closed form of this paper survives with the assorted constants. Three consequences follow, and the first two are exact.

First, the safe face is an exact twin face at every r : the assortment correction is a self-interaction term, and AS and CS are self-identical, earning 59 and causing no harm against themselves exactly as against each other. Neutrality of the face is therefore a property of the payoffs and not of the matching, in agreement with the σ -criterion above.

Second, and for the same reason, *the guard threshold does not move at all*. The comparison behind $L^\dagger$ is between CS and AS at a CAS-rich rest point, and the assortment correction contributes $r[\pi_P(\text{CS}, \text{CS}) - \pi_P(\text{AS}, \text{AS})] = 0$ to its numerator and $r[m(\text{CS}, \text{CS}) - m(\text{AS}, \text{AS})] = 0$ to its denominator, so $L^\dagger = 4.274$ for every $r \in [0, 1]$. The lower edge of the liability valley is immune to population structure.

Third, the upper edge is not. Writing the invasion of AS by CAS in the assorted game gives

$$L_{\text{CAS} \rightarrow \text{AS}}^*(r) = \frac{42.765 - 74.565r}{1 + 8r}, \tag{20}$$

which falls steeply and reaches zero at $r = 0.574$, beyond which unconditional safety needs no liability at all. Because the lower edge is fixed and the upper edge falls, the window narrows monotonically – from a factor of 10.0 at $r = 0$ to 6.5, 4.6, 2.5 and 1.4 at $r = 0.05, 0.10, 0.20$ and 0.30 – and closes at $r = 0.354$. The basin-averaged harm curves in Figure 10C follow: the largest increase of long-run harm along the L -axis falls from 0.299 at $r = 0$ through 0.216, 0.160 and 0.060 to 0.008 at $r = 0.30$, and is exactly zero at $r = 0.40$. Table 7 reports the sweep.

The answer to the question the σ -criterion leaves open is therefore that network reciprocity *shrinks* the liability valley, that it does so entirely from above, and that a modest amount of it is enough: the assortment needed to close the window is smaller than the assortment at which

unconditional safety becomes free, and much larger than the $r = 0.076$ at which conditional safety does. Structure and liability protect the two safe designs in the opposite order, which is the same asymmetry that Theorem 14 reports for liability alone. We state the limits of this: (19) is the mean-field idealisation of structure and does not represent the local invasion geometry of a particular graph, on which clustered aggressors can survive in regions that a well-mixed or assorted calculation does not see.

Table 7: Assortative matching. A design meets a copy of itself with probability r and a random partner otherwise, which is the replicator dynamics of the transformed matrix $\tilde{\pi}(i, j) = r \pi_P(i, i) + (1 - r) \pi_P(i, j)$. The guard threshold does not move at all: the assortment correction is a self-interaction term, and the two designs it compares are self-identical. The upper edge falls, so the window narrows from above and closes at $r = 0.354$; a negative $L_{\text{CAS} \rightarrow \text{CS}}^*$ means conditional safety is protected with no liability at all, which happens from $r = 0.076$.

r	$L^\dagger$	$L_{\text{CAS} \rightarrow \text{AS}}^*$	$L_{\text{CAS} \rightarrow \text{CS}}^*$	width	valley depth
0.00	4.274	42.77	0.551	10.01	0.299
0.05	4.274	27.88	0.182	6.52	0.216
0.10	4.274	19.62	-0.156	4.59	0.160
0.20	4.274	10.71	-0.757	2.51	0.060
0.30	4.274	6.00	-1.274	1.40	0.008
0.40	4.274	3.08	-1.723	closed	0.000

Multi-player faces. Lemma 16 extends to N -player interactions in a weaker but usable form. Let $a_x(s)$ be the payoff to a player using x in a group whose other $N - 1$ members contain s players using j , and let k satisfy the twin hypothesis $a_k(0) = a_i(0)$. At an interior rest point of the $\{i, j\}$ face at frequency p the residents have equal fitness, so the growth rate of a rare k is $\sum_{s \geq 1} \binom{N-1}{s} p^s (1-p)^{N-1-s} [a_k(s) - a_i(s)]$, the $s = 0$ term having vanished. Mixture-freeness therefore survives as a sufficient condition: if $a_k(s) \geq a_i(s)$ for every $s \geq 1$, strictly for some s , then k invades at every interior mixture, and symmetrically for the reverse inequalities. For $N = 2$ the sum has one term and the condition is necessary as well, which recovers Lemma 16.

3.9 What the principal can see

The selection functional charges a principal for the harm its own agent caused, which presumes that incidents are counted without error and booked against the party that caused them. Incident data satisfy neither presumption. We therefore replace the externality intensity by a measured one,

$$\hat{m}(i, j) = \alpha m(i, j) + q m(j, i) + c + \varepsilon_{ij}, \quad (21)$$

where $\alpha \geq 0$ is the fraction of an agent's own incidents that reaches the record, $q \geq 0$ is the weight the principal places on the *counterparty's* record, c is a charge levied irrespective of conduct, and ε is a zero-mean error. Three of the four terms are harmless.

Proposition 20 (Inert measurement error). *Let the charge be $L\hat{m}$ with $\hat{m}$ as in (21).*

- (i) *If $\mathbb{E}[\varepsilon_{ij} \mid i, j] = 0$, then the replicator field, the Nash set and every invasion threshold are exactly those of the noiseless channel.*

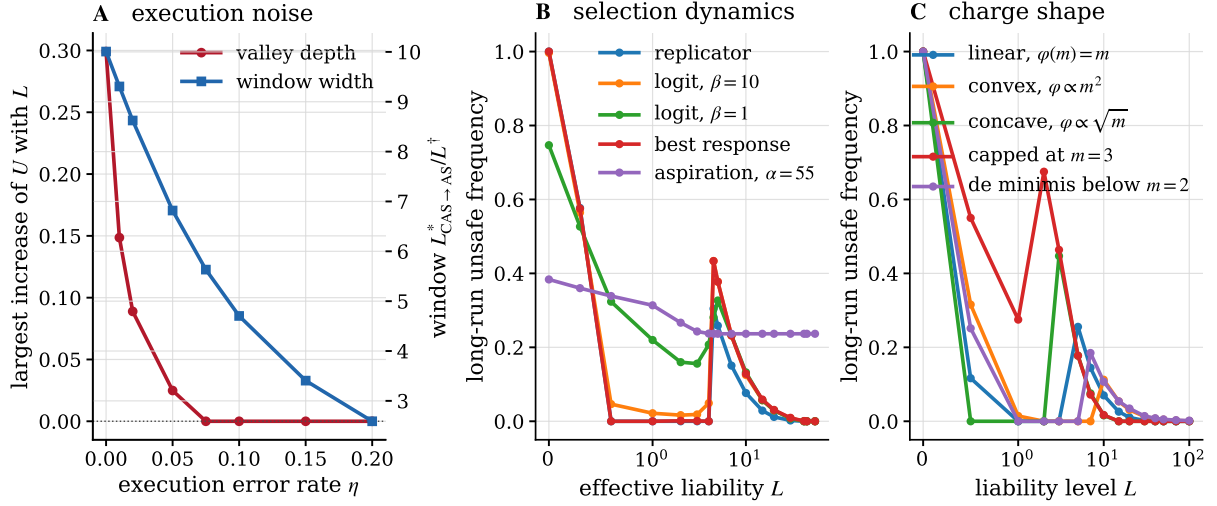


Figure 9: How far the mechanisms travel. **(A)** Execution noise. The bistability window (right axis) persists at every error rate examined, but the largest increase of long-run harm along the L -axis (left axis) vanishes at $\eta \geq 0.075$, because a conditionally safe population stops being safe. **(B)** Long-run harm under five selection dynamics on $\{AS, CS, CAS\}$. The replicator, logit and best-response dynamics all show the valley at the same edge $L^\dagger$, as Proposition 8 requires; aspiration-driven learning does not, but stops responding to liability altogether. **(C)** Long-run harm under five shapes of the liability charge, at equal charge on a design that is unsafe throughout. Every shape has a valley; the concave and capped shapes, which charge the first incident heavily and saturate, have the narrowest.

- (ii) If $q = 0$ and $\alpha = 1$, the same holds for every c , and more generally for every error $\hat{m} - m$ that depends on the counterparty alone.
- (iii) If $q = 0$, the channel maps each threshold L^* to L^*/α and leaves every ratio of thresholds unchanged.

Proof. π_P is affine in the charge base and a design meets the population rather than any single realisation of it, so its fitness depends on the measurement only through $\mathbb{E}\hat{m}$, which is (i). An error that depends on j alone is a column-constant perturbation, so (ii) is Lemma 4. For (iii) the charge is $L\alpha m + Lc$, whose second term is column-constant and whose first is the baseline charge at effective liability $L\alpha$. $\square$

The content of the proposition is that variance is free, and that even bias is free when it is bias about the counterparty rather than about the agent: a principal may misestimate the harm of an interaction by an arbitrary amount without disturbing a single threshold, provided the error does not depend on which design is being charged. Under-reporting is not free but is harmless in a precise sense, since it only relabels the liability axis; a regime that books one incident in three is the regime at one third of the nominal liability, and none of the ratios in this paper move. What is not free is attributing an incident to the wrong party.

Proposition 21 (Misattribution subsidises the first strike). *Set $\alpha = 1 - q$, $c = 0$ and $\varepsilon \equiv 0$ in (21), so that a fixed quantity of blame is split between the two parties to an interaction and a fraction q of it lands on the wrong one. Then*

- (i) *the safe face remains an exact twin face for every q , so Proposition 13 and Theorem 14 apply unchanged;*

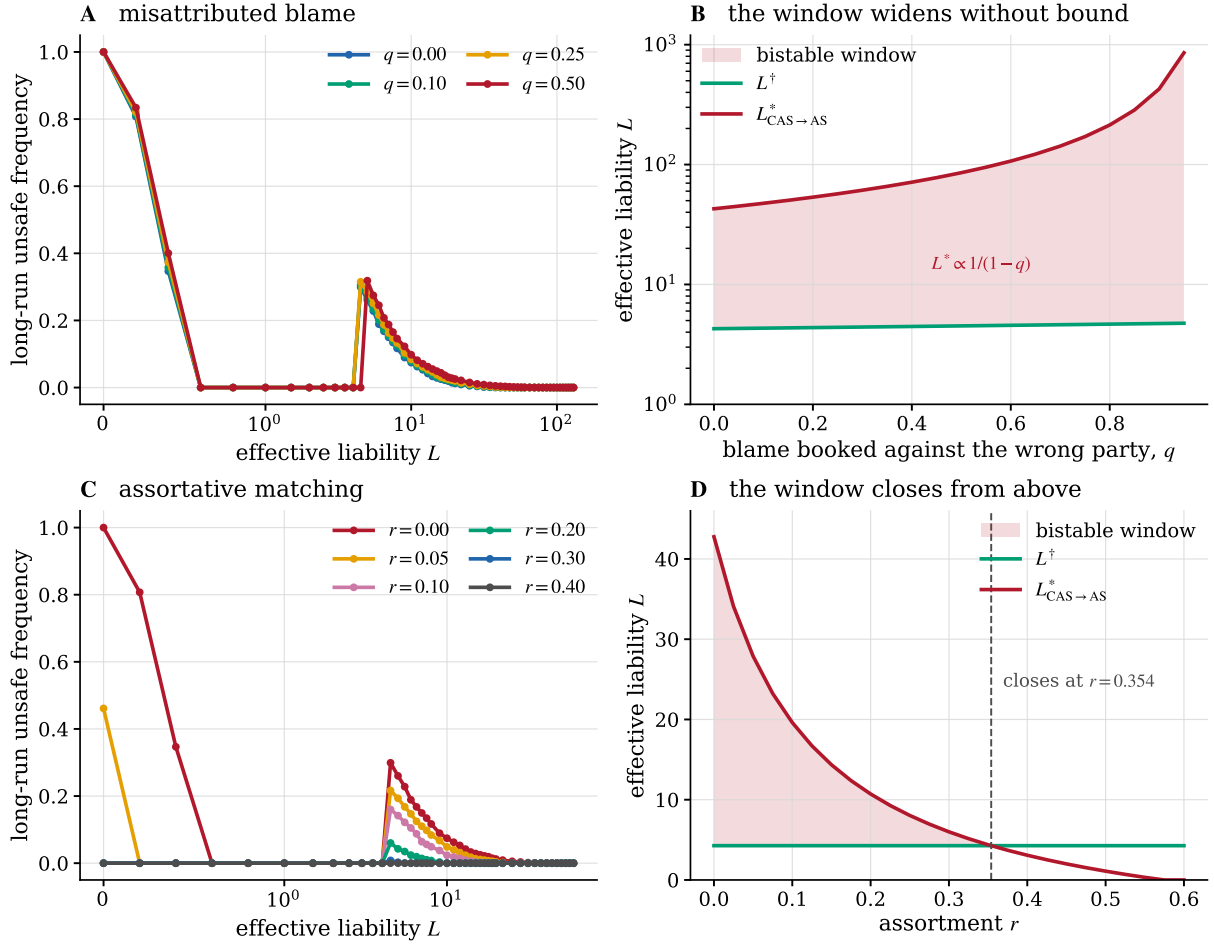


Figure 10: Two features of the environment move the valley, in opposite directions and for the same structural reason: the lower edge of the window is almost inert and the upper edge is not. **(A)** Long-run harm when a fraction q of the blame for an interaction is booked against the wrong party (Section 3.9). **(B)** The two edges as functions of q , log scale. The guard threshold rises by at most 11.8%; the liability protecting unconditional safety diverges as $1/(1-q)$, because the design a first strike exploits has by construction a spotless record, so blame moved onto it is blame destroyed. **(C)** Long-run harm under assortative matching at level r . The dangerous range narrows and is gone by $r = 0.40$. **(D)** The two edges as functions of r . Here $L^\dagger$ is *exactly* independent of assortment, because the assortment correction is a self-interaction term and the two designs the threshold compares are self-identical; the upper edge falls, so the window closes from above at $r = 0.354$.

(ii) $L_{CAS \rightarrow AS}^*(q) = L_{CAS \rightarrow AS}^*(0)/(1-q)$, which diverges as $q \rightarrow 1$;

(iii) $L^\dagger(q) = 18.044/(4.222 - 0.444q)$, which rises from 4.27 to 4.78 over $q \in [0, 1]$, a change of at most 11.8%.

The multiplicative width of the bistable window is therefore

$$\frac{L_{CAS \rightarrow AS}^*(q)}{L^\dagger(q)} = 10.006 \frac{1 - 0.1053q}{1 - q}, \quad (22)$$

which is strictly increasing on $[0, 1)$ and unbounded.

Proof. On $\{AS, CS\}$ both m and its transpose vanish identically, which gives (i). For (ii), the charge difference facing a rare CAS in an AS resident is $(1-q)m(CAS, AS) + qm(AS, CAS) =$

$(1 - q) \cdot 1 + q \cdot 0 = 1 - q$, because unconditional safety never acts unsafely against anybody; the task-payoff difference is untouched, and the threshold is the ratio of the two. For (iii) the same computation at a CAS resident gives $(1 - q) [m(\text{CS}, \text{CAS}) - m(\text{AS}, \text{CAS})] + q [m(\text{CAS}, \text{CS}) - m(\text{CAS}, \text{AS})] = (1 - q) 4.222 + q 3.778$, and (22) is the quotient. $\square$

The asymmetry between (ii) and (iii) is the whole result, and it has a one-line explanation: *the design that a first strike exploits is by construction the design with a spotless record*, so blame moved onto it is blame destroyed, whereas the guard and the aggressor that the lower edge compares are both active against the same opponent and lose roughly the same amount of blame to each other. Misattribution is thus not a symmetric degradation of the instrument. It leaves the protection of conditional safety almost exactly where it was – $L_{\text{CAS} \rightarrow \text{CS}}^*$ moves only from 0.551 to 0.623 across the entire range of q – while making unconditional safety progressively unprotectable, and it widens and deepens the liability valley (Table 8 and Figure 10A,B). At $q = 0.5$, which is what a record that reports the interaction but not the initiator amounts to, the dangerous range has doubled to a factor of 19.0 of liability.

Table 8: Misattribution. A fixed quantity of blame is split between the two parties to an interaction and a fraction q of it lands on the wrong one, so the charge base is $(1 - q) m(i, j) + q m(j, i)$. The guard threshold barely moves, because both designs it compares are active against the same aggressor. The liability protecting unconditional safety diverges as $1/(1 - q)$, because the design a first strike exploits has by construction a spotless record, so blame moved onto it is blame destroyed. The window widens and the valley deepens accordingly.

q	$L^\dagger$	$L_{\text{CAS} \rightarrow \text{AS}}^*$	width	barrier ratio	valley depth
0.00	4.27	42.77	10.01	77.7	0.299
0.10	4.32	47.52	11.00	85.3	0.306
0.25	4.39	57.02	12.99	100.6	0.315
0.50	4.51	85.53	18.96	146.3	0.318

Lagged enforcement. A principal who settles on an incident window that closed τ time units ago charges on the composition as it was, so the fitness of design i is $\sum_j x_j(t) a(i, j) - L \sum_j x_j(t - \tau) m(i, j)$. Every rest point of the undelayed flow is a rest point of the delayed one, because a constant history equals the current state, so *no threshold in this paper moves under lag*: enforcement delay cannot create or destroy a protected regime, it can only change which one is reached. It does change that. Measuring time in units where the largest fitness difference of the selection functional is one, so that τ is comparable with the relaxation time of the flow, the share of random starts that end at the unsafe attractor at $L = 6$ rises from 0.58 at $\tau = 0$ to 0.79 at $\tau = 20$, while at $L = 10$ it falls from 0.54 to 0.29. The effect is a reallocation of basins of up to about twenty percentage points, in a direction that depends on where in the window the liability sits, and it does not accumulate into a change of regime.

3.10 An eco-evolutionary ratchet

Deployed capability does not disappear. We couple the deployment dynamics to a state $z \in [0, 1]$ measuring the stock of diffused unsafe capability, which enters through enforcement: as a

capability proliferates, attributing harm to a particular principal becomes harder, so the liability that actually reaches a principal decays. The coupled system is

$$\dot{x} = \text{Rep}(x; a - L_{\text{eff}}(L, z)m), \quad \dot{z} = \varepsilon U(x)(1 - z), \quad (23)$$

with $\varepsilon \ll 1$ separating the time scales. The construction differs from renewable game–environment models [30, 31] in that $\dot{z} \geq 0$ identically: the environmental state is a ratchet.

We do not assume a functional form for the erosion. An *erosion channel* is any $L_{\text{eff}} : \mathbb{R}_{\geq 0} \times [0, 1] \rightarrow \mathbb{R}_{\geq 0}$ that is continuous, non-increasing in z , strictly increasing in L , and normalised by $L_{\text{eff}}(L, 0) = L$. Write

$$\rho := \frac{L_{\text{eff}}(L, 1)}{L} \in [0, 1) \quad (24)$$

for the fraction of enforcement that survives full diffusion; for a channel that is homogeneous in L , as both examples below are, ρ does not depend on L .

Theorem 22 (Hysteresis width). *Let the safe attractor have $U = 0$ exactly, as in Proposition 13, let L_{eff} be any erosion channel with residual fraction ρ , and consider a quasi-static sweep of L that starts in the protected regime with $z = 0$. The protected regime is lost at $L_{\downarrow} = L_{\text{CAS} \rightarrow \text{CS}}^*$, after which $z \rightarrow 1$, and is recovered on the return sweep at $L_{\uparrow} = L_{\text{CAS} \rightarrow \text{CS}}^*/\rho$. Hence*

$$\frac{L_{\uparrow}}{L_{\downarrow}} = \frac{1}{\rho} = \begin{cases} \frac{1}{1 - \theta}, & L_{\text{eff}} = L(1 - \theta z), \\ 1 + \kappa, & L_{\text{eff}} = L/(1 + \kappa z), \end{cases} \quad (25)$$

for every $\varepsilon > 0$ and for every channel with that ρ .

Proof. On the protected branch $U(x) = 0$, so the second equation of (23) gives $\dot{z} = 0$ for every ε ; hence z stays at its initial value 0 and $L_{\text{eff}} = L$ by the normalisation. The branch is lost exactly when CAS can invade the resident safe face, i.e. at $L = L_{\text{CAS} \rightarrow \text{CS}}^*$ by Lemma 7 and (8), and this is a condition on π_P alone, so by Proposition 8 it does not depend on the imitation rule. Once lost, $U > 0$ on the attractor, so $\dot{z} > 0$ until $z = 1$; since $\dot{z} \geq 0$ always, z cannot return. On the return sweep the population therefore faces $L_{\text{eff}}(L, 1) = \rho L$, and recovery requires $\rho L > L_{\text{CAS} \rightarrow \text{CS}}^*$. Both thresholds are conditions on L_{eff} evaluated at the two endpoints $z \in \{0, 1\}$; the channel enters only through ρ and ε enters nowhere, since it scales the speed of z but not its limit. $\square$

The non-trivial ingredient is neither the erosion channel nor the time-scale separation. It is Proposition 13: because the protected branch has $U = 0$ *exactly*, the ratchet variable is frozen there, so the loss threshold is the bare invasion threshold and the shape of the channel between its endpoints is irrelevant. Everything the channel contributes is the single number ρ , and two channels sharing a ρ share a width, as (25) shows for $\theta = 0.9$ and $\kappa = 9$.

Figure 11 confirms the prediction numerically at $\theta = 0.9$: the descending branch loses protection between $L = 0.62$ and $L = 0.54$, bracketing $L_{\text{CAS} \rightarrow \text{CS}}^* = 0.551$, and the ascending branch recovers it at $L = 5.55$ against the predicted 5.51. The loop width is the predicted factor of ten. Table 9 repeats the sweep over both channels, two residual fractions and a factor of twenty in ε ; the observed ratio tracks $1/\rho$ in every case and is unchanged by ε , as the theorem requires.

The one assumption that does bind is exactness of $U = 0$. If the protected branch carries any residual harm – as it does under execution noise, where a conditionally safe population retaliates

Table 9: Hysteresis width across erosion channels and diffusion speeds. The quasi-static sweep starts in the protected regime with $z = 0$; $L_\downarrow$ and $L_\uparrow$ are the last grid point at which protection holds on the descending branch and the first at which it is recovered on the ascending one, on a geometric grid whose spacing is a factor of 1.08. The predicted loss threshold is $L_{\text{CAS} \rightarrow \text{CS}}^* = 0.551$ throughout. Changing ε by a factor of twenty leaves the loop unchanged, and the two channels agree wherever they share a residual fraction ρ .

channel	ε	ρ	$L_\downarrow$	$L_\uparrow$	observed	$1/\rho$
$L(1 - \theta z)$, $\theta = 0.9$	0.05	0.10	0.564	5.747	10.19	10.00
$L(1 - \theta z)$, $\theta = 0.9$	0.01	0.10	0.522	5.747	11.01	10.00
$L(1 - \theta z)$, $\theta = 0.9$	0.20	0.10	0.564	5.747	10.19	10.00
$L(1 - \theta z)$, $\theta = 0.5$	0.05	0.50	0.522	1.132	2.17	2.00
$L/(1 + \kappa z)$, $\kappa = 9$	0.05	0.10	0.609	5.747	9.43	10.00
$L/(1 + \kappa z)$, $\kappa = 1$	0.05	0.50	0.564	1.132	2.01	2.00

against its own trembles (Section 3.7) – then $\dot{z} > 0$ on that branch too, and the protected regime is not frozen but slowly creeping. The loss threshold then depends on ε , since what matters is whether the sweep is completed before z has accumulated. That is the honest boundary of the result: the hysteresis width is exactly computable when safety is exact, and becomes a race between the sweep and the ratchet when it is not.

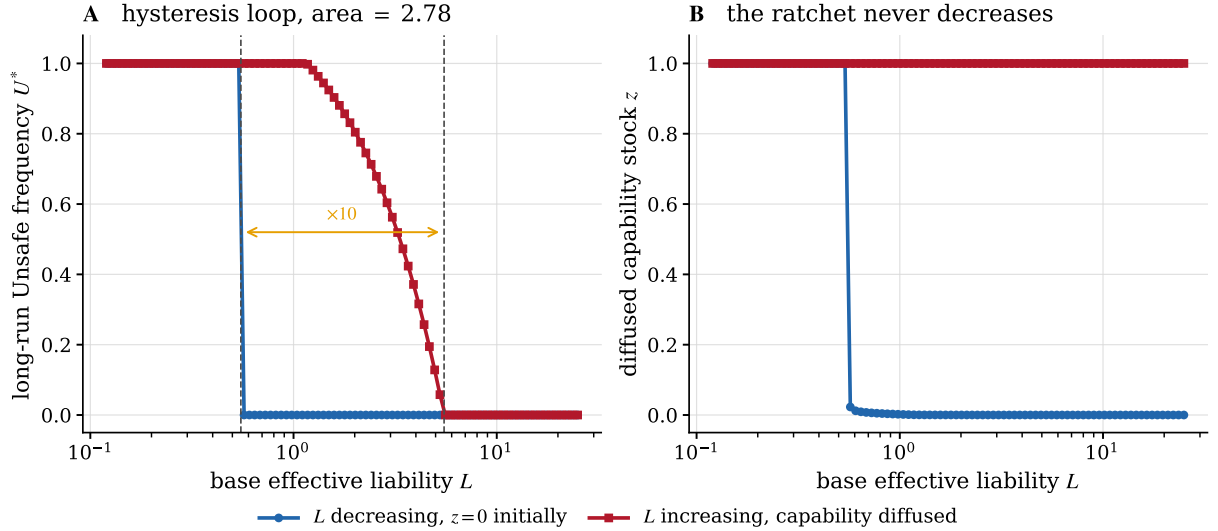


Figure 11: Irreversible capability diffusion produces hysteresis. (A) Quasi-static sweep of the base effective liability, decreasing from a protected state and then increasing again. (B) The ratchet variable, which never decreases. At $\theta = 0.9$ the recovery threshold is ten times the loss threshold, as Theorem 22 predicts.

The practical content is a timing asymmetry. Because $L_\uparrow$ exceeds $L_\downarrow$ by $1/\rho$, the cost of preventing a transition is smaller than the cost of reversing one by exactly the factor by which enforcement erodes, and the required overshoot is known in closed form once ρ is estimated.

4 Discussion

What the model says and does not say. The results are statements about a specific stylised race with four deterministic designs. They are not predictions about frontier AI systems, and the external harm h has no calibration, which is why every result is stated on the whole L -axis and normalised by the prize. What does transfer is the structure. Whenever the entity that selects a design is not the entity that plays the game, and internalises only part of the harm, the population dynamics is the replicator equation of the selector’s functional. The four mechanisms identified here – dominance of conditional over unconditional aggression, neutrality of the safe face, taxation of retaliation, and irreversibility – then follow from qualitative features of the payoff structure rather than from the particular numbers. The first requires only that a conditional design imitate the unconditional one against unsafe opponents. The second requires only that safe designs be behaviourally indistinguishable against safe opponents. The third, by Lemma 16, requires only that the second hold and that retaliation be classified as harmful by the liability rule.

Evaluations measure the wrong object. Corollary 11 says that a filter defined on solo behaviour removes a weakly dominated design. That statement is a formal counterpart of a known practical worry, that a model may behave acceptably when probed and unacceptably in deployment [11, 44]. The mechanism here requires no deception: CAS is not hiding anything. It is simply conditional, and an unconditional probe cannot elicit the condition. The corrective suggested by the model is not stricter thresholds but a different measurement, and Proposition 12 makes it precise: probe the design against a *reciprocating* counterpart. That single change widens the margin between conditional safety and conditional aggression from 0.132 to 0.539, and it survives execution noise better than the solo probe does. Probing against a maximally hostile counterpart instead is counterproductive, because a conditionally safe design retaliates against it in 87% of rounds and is scored as unsafe. Panel B of Figure 3 is the underlying argument – the two orderings disagree on exactly one design, and it is the one that governs the equilibrium.

Stated as a protocol for the agents this model is about, the recommendation is short. Run the candidate against a scripted counterpart that opens safe and thereafter mirrors the candidate’s previous action; score the candidate only on the rounds that are not preceded by an unsafe action of the counterpart; and admit on that score rather than on behaviour against a uniformly safe environment. The scripted counterpart needs no capability of its own, which matters because an evaluation that required a second frontier system to play the aggressor would not be run. Two properties make the protocol cheap. The margin it has to resolve is four times the margin of the solo probe, so at the same confidence it needs $n \geq 7$ episodes rather than $n \geq 111$, and it still needs only $n \geq 16$ at a 5% execution error rate. And the probe must not be hostile: a counterpart that opens unsafe destroys the separation, because the design one wants to admit is the design that fights back. The tension this creates with liability is exact rather than rhetorical. An evaluation rule that scores retaliation as unsafe and a liability rule that charges retaliation as harm are the same mistake made at two different layers, and Corollary 18 is what it costs when both are made at once.

Do not mandate unconditional safety. Theorem 14 inverts a natural intuition. Unconditional safety looks like the strictest possible design, and it is the most fragile: it demands the

highest liability of any safe composition, by a factor of 77.7 at the baseline and up to 90.4 at large prizes. Conditional safety is cheap to protect because it makes exploiting it unprofitable, and at high private risk it needs no liability at all. Because the safe face is neutral, a policy that fixes the composition is not self-sustaining; it must be maintained against drift. Two of the robustness checks strengthen this conclusion rather than qualifying it. Under execution noise the face stops being neutral and the gradient points at conditional safety throughout the policy-relevant range (Section 3.7), and in a structured population conditional safety needs no liability at all once assortment exceeds $\sigma = 1.10$ (Section 3.8). The one regime in which unconditional safety is selected for is high liability, which is also the regime in which it is cheapest to protect – the instrument and the composition it induces work against each other at every level below that.

Liability has a valley, and it is wide. Corollary 18 is the result most likely to matter outside the model. Liability instruments are usually assumed to be monotone: more internalisation, less harm. Here an entire decade of L is worse than the lower end of the range, because the instrument does not distinguish unsafe behaviour that initiates from unsafe behaviour that retaliates. Theorem 17 locates the boundary exactly, and Section 3.6 shows that a window of this kind exists across the whole range of prizes and risk levels we examined. It also survives every enrichment of the design pool, every imitative selection dynamics and every shape of charge that we tested, and it closes only when execution error exceeds about 7.5% – at which point the protected state has itself stopped being safe (Sections 3.7 and 3.8). What moves is the location of the window, by a factor of three between the reduced pool and the enlarged one, so the model identifies the existence of a dangerous range and not its address.

Structured populations. The dynamics is well mixed, but two of the four mechanisms can be carried to structured populations exactly. Network reciprocity generally favours conditionally cooperative strategies by clustering them [45, 46], and the structure coefficient of Tarnita et al. [41] makes that quantitative here. Equation (18) shows that the safe face is neutral for every structure, and that assortment substitutes for liability at a steep rate: protecting unconditional safety needs $L > 6.14$ at $\sigma = 1$ but nothing at all at $\sigma \geq 2.93$, and protecting conditional safety needs nothing at all already at $\sigma \geq 1.10$. The weak dominance of CAS over AU (Theorem 10) is a pointwise payoff comparison and survives any matching structure, though Section 3.7 shows it does not survive an enlarged design pool. The valley the structure coefficient cannot reach, because it orders each pair of designs and a bistable pair has no order; assortative matching can, since it keeps the dynamics, and the answer is that structure shrinks the valley. Equation (20) shows the shrinking is entirely from above: the guard threshold is exactly independent of assortment, because the assortment correction is a self-interaction term and the two designs the threshold compares are self-identical, while the liability that protects unconditional safety falls to zero at $r = 0.574$. The window closes at $r = 0.354$. The policy reading is that structure and liability are substitutes with very different shapes – an ecosystem in which agents mostly interact with their own kind needs less liability *and* has a narrower dangerous range – and that interoperability mandates, which lower r , therefore make the choice of liability level more consequential rather than less. What remains open is the local invasion geometry of a specific graph, which the mean-field transform (19) does not represent and on which clustered aggressors can persist in regions an assorted calculation does not see.

Reading the liability axis. We do not calibrate h , and no responsible calibration is available: the external harm of one unsafe development step in a real deployment race is not a measured quantity. The axis can nevertheless be read in units of the prize, which is the one quantity a deploying firm does know. In those units the four thresholds of (9) are $L/B = 0.0012$, 0.0055 , 0.043 and 0.428 , and Section 3.6 shows that the outer one stays between 0.38 and 0.61 across the whole grid of prizes and risk levels. The statement that survives is therefore a comparative one: protecting unconditional safety costs about half the prize per unsafe step, protecting conditional safety costs under one per cent of it, and the dangerous range lies between roughly four per cent and half of the prize. A regulator who can estimate what a deployment race is worth to its participants – a procurement contract, a market position, a first-mover advantage – can locate a candidate regime on that axis without ever estimating h . Doing so honestly for a named instrument requires an empirical estimate of the payoff and harm matrices of real agent designs, which is the first of the two extensions in Section 5, not something the present model supplies.

Instruments that do not tax retaliation. Proposition 6 says that a charge indexed on the counterparty’s behaviour is column-constant and therefore dynamically free, and Corollary 18 says that the valley exists because the actual instrument cannot tell initiation from retaliation. Three implementable approximations follow, in increasing order of information required. The cheapest is to shape the charge rather than to index it: Table 6 shows that a concave or capped schedule, which charges the first incident heavily and saturates, narrows the window to 4.9 from the linear 10.0 , because saturation spares the retaliator, whose incident count is high, more than it spares the initiator, whose count is one. Deductible-and-cap structures in insurance have exactly this shape, which is a reason to prefer liability that is mediated by insurers to liability that is assessed per incident. The second is to condition the charge on an observable proxy for who moved first – an incident report that records the counterparty’s prior action, which is what an audit log of an agent interaction already contains. The third is certification of the response rule itself rather than of the behaviour it generates [47]: a design certified as retaliation-only is charged as if column-constant, which by Lemma 4 leaves the dynamics untouched. All three are information problems rather than legal ones, which is the useful reframing: the obstacle to a non-distorting liability rule is that incident data does not currently record who initiated.

Section 3.9 turns that reframing into a quantitative requirement, and in doing so tells a regulator which parts of a reporting regime are worth paying for. Precision is not one of them. Proposition 20 says that zero-mean error in counting incidents changes nothing at all, that a flat charge per deployment changes nothing at all, that any error depending only on the counterparty changes nothing at all, and that systematic under-reporting merely relabels the liability axis, so a regime that books one incident in three is the regime at one third of the nominal liability. Attribution is the one thing worth paying for: by Proposition 21 the liability that protects unconditional safety scales as $1/(1 - q)$ in the fraction q of blame booked against the wrong party, and at $q = 0.5$ – a record that establishes an interaction went wrong but not who started it – the dangerous range has doubled. The instruction to a designer of an incident-reporting standard is therefore to spend the budget on identifying the initiator and not on measuring severity, which inverts the usual emphasis.

One caveat belongs with the counterparty-indexed charge that Proposition 6 recommends, and it is the reason we call it an approximation rather than a solution. A charge that depends on the counterparty is inert on the *play* dynamics precisely because it does not depend on what

the charged agent does; but it does depend on whom the agent meets, and in any ecosystem where partners are chosen rather than drawn, that dependence is an incentive to choose them. A principal facing a column-constant charge has no reason to change its design and every reason to route its agent towards counterparties with clean records, which is a matching game our model does not contain and which would erode the very assortment Section 3.8 shows to be protective. The inertness result should therefore be read as a statement about one layer: counterparty indexing moves the distortion out of the design margin and into the matching margin. Whether that is an improvement depends on which margin the regulator can observe.

Limitations. Five are material, and Sections 3.7–3.8 bound three of them rather than removing them. The interaction layer has two players, so entry of new designs is modelled only as mutation; enriching the pool leaves every mechanism qualitatively intact except weak dominance, which fails once a design punishes for longer than it is punished. The valley is a low-error phenomenon, closing at execution error rates above about 7.5%; whether real agent ecosystems sit below that is an empirical question we cannot answer. The principals are homogeneous and passive: they select on π_P but do not strategise about each other, which rules out the enforcement-avoidance behaviour a richer model would include. The ratchet channel is assumed rather than derived; Theorem 22 now isolates exactly what does not depend on that assumption – everything except the residual fraction ρ – but the assumption of erosion itself remains, and the freezing of the protected branch fails under execution noise. Finally, the analysis is deterministic in h and λ , whereas real liability arrives with probability and delay. Section 3.9 removes part of that limitation and sharpens what is left of it: the probabilistic part is harmless, since zero-mean error and systematic under-reporting are absorbed exactly, and the delay is harmless for the thresholds though not for the basins, but the analysis assumes the charge is levied on an incident record whose attribution error q is exogenous and known. In a real regime, q is itself chosen by parties with an interest in its value, and a model in which principals contest attribution rather than accept it is the natural next step. The non-linear charges of Section 3.8 address the shape of the schedule but not its stochasticity.

5 Conclusion

Separating the payoff that an AI design earns from the payoff that decides whether it is replicated changes the dynamical system, not merely its parameters. Working through that separation in the reduced AI race yields four results that are invisible when the two payoffs are identified. Pre-deployment solo evaluation cannot change the equilibrium, because it removes a weakly dominated design. The invasion barrier protecting a safe population is controlled by a neutrally drifting composition, and falls by a factor of 77.7 along it. Long-run harm is non-monotone in liability, with a bistable window that exists at every prize and risk level we examined and traps the population for up to 4×10^{13} generations. And irreversible capability diffusion makes the loss of a protected regime cost a known factor less than its recovery.

Each result points at the same target. Governing an ecosystem of deployed agents by acting on the designs – filtering them, training them, mandating them – addresses the layer that does not carry the selection gradient. Acting on the functional that principals select by does carry it, but the instrument is not monotone, and the range in which it backfires is precisely the range in which it is strong enough to punish retaliation and too weak to protect unconditional safety.

Locating that range for a given deployment setting seems to us more useful than arguing about whether liability should be strengthened.

Two extensions follow directly. Estimating the payoff and harm matrices of real agent designs empirically, rather than from a stylised race, would place an actual deployment ecosystem on the L -axis. And enriching the principals – letting them anticipate liability, compete, or share it – turns the selection functional itself into a strategic object, which is the natural second layer of the problem.

Declarations

Code and data availability. All code, tables and figures are reproducible from a public repository released under the MIT licence, <https://github.com/trungkiet2005/deployment-layer-selection>. Evolutionary computations use EGTtools [38].

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A Invasion thresholds

Table 10 applies Lemma 7 to every ordered pair of designs in Tables 1 and 2. Each row is the pair $(\delta a, \delta m)$ of that invasion together with the critical effective liability $L^* = \delta a / \delta m$ and the side of L^* on which the invader grows; the four rows that organise the paper are collected in (9).

Table 10: Closed-form critical effective liabilities $L^* = \delta a / \delta m$ for every ordered invasion at $p_r^{\max} = 0.6$. The last column states when the invader has a positive growth rate.

Invader	Resident	δa	δm	L^*	invades
AS	AU	-21.800	-9.000	2.422	$L > L^*$
AS	CS	0.000	0.000	–	–
AS	CAS	-18.600	-9.000	2.067	$L > L^*$
AU	AS	-10.360	9.000	-1.151	$L < L^*$
AU	CS	-11.640	9.000	-1.293	$L < L^*$
AU	CAS	0.000	0.000	–	–
CS	AS	0.000	0.000	–	–
CS	AU	-10.600	-1.000	10.600	$L > L^*$
CS	CAS	-0.556	-4.778	0.116	$L > L^*$
CAS	AS	42.765	1.000	42.765	$L < L^*$
CAS	AU	0.000	0.000	–	–
CAS	CS	2.631	4.778	0.551	$L < L^*$

B The other risk treatments

The source experiment manipulates the maximum private setback risk $p_r^{\max} \in \{0.1, 0.6, 0.9\}$ in (2). The unsafe counts $m(i, j)$ are identical across treatments, because $p_r^{\max}$ scales the realised payoff but leaves the stage matrix (1) and hence the action paths untouched; only $a(i, j)$ changes. Evaluating the interaction game without liability ($L = 0$) reproduces the qualitative ordering reported in the source study [18]: the neutrally stable set is $\{\text{AU}, \text{CAS}\}$ at $p_r^{\max} = 0.1$ and 0.6 , and $\{\text{CS}\}$ at $p_r^{\max} = 0.9$, so high private risk favours conditional safety even before any external liability is applied. Our main analysis uses $p_r^{\max} = 0.6$, the intermediate treatment, because it is the one in which the conditional designs are both present and neither is trivially dominant; Section 3.6 reports every threshold for the other treatments.

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